SINGLEHEAD AUTOMATIC TURRET		
Model		
EAGLE VA TURRET		
Serial number		
A2055		
Manufacturing year	*Update*	
2007	**A**	**02–07**
CLOSYS S.r.l. – 14053 CANELLI (AT) – ITALY		

NOTICE - EDUCATIONAL USE ONLY

CLOSYS

Table of contents

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TECHNICAL DATA OF THE MACHINE

Model:	EAGLE VA TURRET
Serial number:	A2055
Rotation:	CLOCKWISE
Manufacturing year:	2007

a. Weight

Weight of the machine	500	kg

b. Production capacity

Pieces/hour	3.000

c. Type of closure

Type of bottle	Equipment identification*	
glass 500 ml.		

Type of cap	Equipment identification*
Aluminium screw cap	

*** in case of several formats.**

d. Electrical data.

Feeding voltage:		
main	380	V
	50	Hz
auxiliary	24	V
	50	Hz
Installed power:		
head rotation motor	0.37	kW
caps sorter motor	0.37	kW
Installed total power:	**0.74**	kW
Maximum absorption:	2.00	A

e. Air data.

Sterile air capacity	400 ≃	Nl/min
Sterile air minimum pressure	6	bar
Sterile air maximum pressure	8	bar

A CLOSURE PROCESS AND PROPER USE

Use the machine, exclusively, following the instructions contained in this manual. Any other non mentioned use, it must be considered "improper use"

The EAGLE VA turret is able to:

- apply aluminium caps to thread.

This model applies caps on bottles as described in table at page 1 point "**c.**".

Caps arriving from the caps sorter (**A**), are driven by the caps chute (**B**) until the distribution head (**C**).

Cap placed about 45°, is taken with a system called "pick off", by the bottle that is coming during the rotation.

The star−wheel rotation (**D**) transfers bottles under the capping head (**E**) where the closure of the cap on the bottle takes place

The capping head following the profile of a cam will thread and seam the cap on the bottle by the rotation of threader and seamer rollers.

After the closure, there is the detachement from the bottle in ext and the rearmement for the following closure

Fig.1

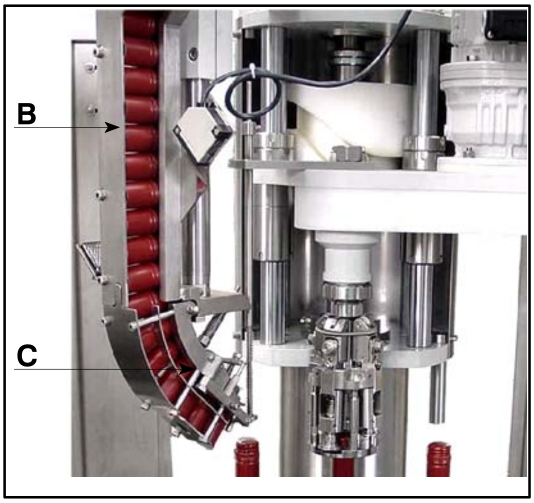

Fig.2

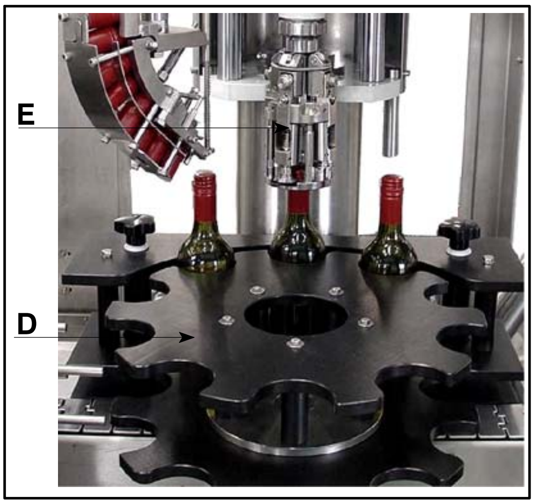

Fig.3

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B

GENERAL INFORMATION

1. MANUFACTURER DECLARATION

The original of the MANUFACTURER DECLARATION is delivered together with the machine documents and must be kept till the machine is scrapped.

2. MACHINE IDENTIFICATION

The machine is identified by the CE marking according to the specifications of the European Directive 98/37/EC (see par. page 13).

The identification plate is on the machine rear side, on the base.

The identification plate is on the upper side of the turret.

To order spare parts or to contact the manufacturer (communications – info request) always indicate:

- Machine model (**MODEL**)
- Machine serial number (**SERIAL N.**)
- Manufacturing year (**YEAR**)

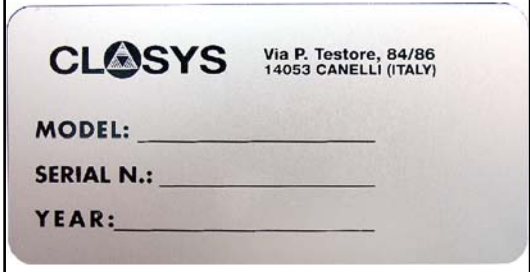

Fig.4

3. FOREWORD

This manual is addressed to the operators and specialized staff to enable a correct machine use.

The operator finds here instructions and indications for:

- a correct machine installation
- a functional description of the machine and of each component, including also the accident–preventing safety norms
- adjustments on the machine while setting and starting it up
- a correct scheduled and routine maintenance.

The operator can thus know the machine features and operation.

The used terms and symbols are specified here below to understand the manual fully:

TERM	***DESCRIPTION***
OPERATOR	person charged with the machine operation, adjustment, routine maintenance and cleaning.
QUALIFIED TECHNICIAN	specialized person, suitably trained and authorized for installation, extraordinary maintenance or repair interventions requiring a special machine knowledge.
EMPLOYER (OR CUSTOMER)	person legally responsible for the machine use.

GRAPHIC SYMBOL	***DESCRIPTION***
	Manual section reserved to the *qualified technician*
	Pay attention to the accident–preventing norms
	Possibility of damaging the machine and/or its components
	Special note

Before any operation with or on the machine read the procedures and warnings herein carefully.

SOME FIGURES IN THIS MANUAL ARE APPROXIMATE AND MAY DIFFER AS FOR THEIR ASPECT FROM THE MACHINE DESCRIBED IN THIS MANUAL. Such differences represent machine components that can be positioned differently depending on the version, but that can be easily recognized. The reported instructions and use modes for all described components anyway remain valid.

4. KEEPING THIS USE INSTRUCTION MANUAL

To keep the "Manual" correctly, it is recommended to:

- keep the manual in the room where the machine operates and in areas protected against humidity, so as not to jeopardize its life in time;
- use the manual not damaging it;
- not to remove, add, modify or edit any part of the document; possible updates shall be made exclusively by *CLOSYS S.r.l.*
- give the manual to the other owner of the machine.

5. REPRODUCTION RESTRICTION, RESPONSIBILITY AND COPYRIGHT

Copyright by *CLOSYS S.r.l.*

The reproduction of this manual is allowed only for training and information purposes of the staff charged with the machine use.

No reproduction allowed for other purposes, without explicit authorization by *CLOSYS S.r.l.*

<u>As *CLOSYS S.r.l.* is steadily involved in the improvement of its products, the information in this manual may be changed without notice. Utmost care was used while preparing this manual. Anyway, *CLOSYS S.r.l.* does not undertake any responsibility for errors or omissions. No responsibility is taken for damages resulting from the use of the information herein.</u>

6. REVISIONS

The manufacturer engages to issue future revisions of the manual following to modifications on the machine.

7. CHARACTERISTICS OF THE OPERATORS

To understand the instructions (text and figures) the machine operators must have (or get, though a suitable training and education) at least the following characteristics:

- sufficient general and technical knowledge to read and understand the manual content in the sections concerning them and to interpret drawings and schemes correctly;
- capacity of understanding and interpreting symbols and pictograms;
- knowledge of the main accident–preventing and hygienic norms;
- global knowledge of the machine and of the environment where the machine is installed;
- capacity of behaving correctly in case of machine emergency situations.

The qualified technicians, besides the above characteristics, must also have a good technical preparation and/or a suitable working experience in the relevant field. Further, they must have specific and special (mechanical and electrical) technical knowledge needed for the operations described in this manual.

8. CUSTOMER'S OBLIGATIONS

The customer shall suitably train and inform the operators on the following topics concerning safety while using the machine:

- specific instructions for the machine use and maintenance.
- general accident–preventing norms, or norms specified by international directives and by laws of the machine destination country.

After the training, the customer must ensure the qualified technicians and operators have understood the above points, and must make them sign the "Performed training sheet" enclosed herewith.

These qualified technicians and operators only can work on the machine.

The customer, or a persons charged by him, shall deliver the P.P.M. (Personal Protection Means) required for all operations on the machine needing them (maintenance, cleaning, etc.).

8.1. INFORMATION

The following requirements are taken from the European Directive 89/391/EEC and following modifications on the improvement of the workers' safety and health in the working environment.

General information

The customer must inform the worker on:

1) risks for safety and health connected with the company activity in general;
2) used protection and prevention measures and activities;
3) specific risks he is exposed to depending on the performed activity, safety norms and company arrangements on this matter;
4) hazards connected with the use of dangerous substances and compounds on the basis of the safety data sheets specified by the norms in force and by the good practice norms.

Use of the work equipment

1) The customer arranges for the charged workers to have all information and use instructions needed for safety reasons for each work equipment they use and concerning:
   - the equipment use conditions also on the basis of the conclusions from previous use experiences on the work equipment;
   - expectable unusual situations.
2) The use instructions and information must be easily understandable by the concerned workers.

P.P.M. use

The customer:

- gives understandable instructions to the workers,
- preliminarily informs the worker on the risks he is protected from through the P.P.M.,

8.2. JOB INSTRUCTIONS

General information

1) The customer ensures every worker receives suitable and sufficient job instructions on safety and health, with special reference to his working station and tasks.
2) The training must be performed:
   - when the worker is employed;
   - whenever he is moved or his tasks are changed;
   - when new work equipment or new technologies are adopted, as well as new hazardous substances and compounds are used.
3) Training must be periodically repeated according to the risk evolution, namely when new risks show up.

Use of the work equipment

The customer makes sure that:

- the workers charged with the use of work equipment receive a suitable training accordingly;
- the workers charged with the use of equipment requiring special knowledge and responsibility receive a suitable and specific training, so that they can use such equipment safely and suitably, also as far as risks caused to other people are concerned.

P.P.M. use

The customer:

- ensures a suitable training and arranges, if necessary, a special training on the correct and practical use of the P.P.M.

8.3. TRAINING

Training is the activity aimed at understanding the correct machine use, thus complying with the obligations described in the previously mentioned European Directive.

9. MEASURES BY THE CUSTOMER

Excluding special contract conditions, the customer must arrange the following:

- pneumatic supply
- generic tools for the machine maintenance and consumables
- lifting means suitable for the machine handling.

10. HOW TO ASK FOR INTERVENTIONS

CLOSYS S.r.l. puts its own Technical Service at the disposal of its customers to solve any problem regarding the machine use and maintenance.

The interventions must be required after carefully evaluating the faults and their reasons, and stating:

- the details on the occurred faults
- the performed checks
- the performed adjustments and resulting effects
- any further information deemed useful.

The requests for interventions by the Customer Technical Service must be forwarded to the following address:

CLOSYS S.r.l.

Via P. Testore, no 84/86 – 14053 CANELLI (Asti) ITALY

Phone: +39 0141 822062

Fax: +39 0141 824380

Web site: http://www.closys.it

11. HOW TO ORDER SPARE PARTS

All orders for spare parts must be sent to *CLOSYS S.r.l.*

The warranty is cancelled if non–genuine spare parts are installed.

To order spare parts, fill in the suitable form in the ”*SPARE PART CATALOGUE*” for the machine.

Spare part replacement must be carried out by specialized technicians following the procedures and using ALL SAFETY PRECAUTIONS described in the chapters ”Maintenance” and ”Safety”.

12. WARRANTY AND FINAL TEST

The machine is delivered to the customer ready to be installed, after having passed all tests arranged by the manufacturer at the factory, according to the laws in force. During the warranty, the manufacturer engages to eliminate possible faults or defects provided that the machine has been used correctly, in compliance with the instructions of the use and maintenance manuals.

For everything not specifically indicated in this manual, refer to the "*GENERAL SALES CONDITIONS point 8 WARRANTY – LIABILITY RESTRICTIONS:*"

***CLOSYS S.r.l.* shall not be liable for inconveniences, breakage, accidents, etc., due to the non–knowledge (or anyway to the failed application) of the specifications herein. The same applies to changes, modifications and/or installation of non–genuine accessories or spare parts without previous authorization.**

In particular, *CLOSYS S.r.l.* declines all liabilities for damage due to:

- acts of God;
- wrong maneuvers;
- failed maintenance;
- damage of the electric or electronic components due to condensate or contact with outer conductors.

***CLOSYS S.r.l.* further declines all liabilities resulting from an incorrect or improper machine use.**

<u>The warranty shall be cancelled if machine components are modified or re-placed, or if the software of programmable components is varied, without previous authorization by the manufacturer.</u>

13. NORMS – CERTIFICATIONS – USE RESTRICTIONS

13.1. REFERENCE TECHNICAL NORMS

The following norms have been referred to while designing and manufacturing the machine:

EC Directive

Directive 98/37 EC.

Directive 89/391 EEC and following modifications and additions acknowledged up to today.

Directive 73/23 EEC.

Directive 89/336 EEC.

13.2. EXPLOSIVE ATMOSPHERE

The machine is not suitable for being used in premises where atmospheres with explosion risk are expected or expectable. The customer must not use the machine in explosive or partially explosive atmosphere.

13.3. LIGHTING

The machine does not feature a own lighting, as it is expected to operate in standard lighted premises.

13.4. VIBRATIONS

The machine complies with the limits set by the Directive 2002/44/EC.

14. PRECAUTIONS FOR THE OPERATORS' SAFETY

When glass container parts must be removed from the machine it is necessary: to turn the main switch off, to wear protective gloves with a min. cutting protection 2 and to remove each foreign matter using a small broom and a dustpan.

To install, maintain or change size, sometimes it is necessary to change the cap/cork guide or the delivery chute. Due to the weight of these components and to their positioning in the machine upper part, this operation must be carried out with the machine off, using suitable lifting systems.

15. SAFETY NORMS FOR THE OPERATION

Wrenches and tools used for the maintenance or the access to machine hazardous sections must not be left at the disposal of operators not suitably trained for the size change or the maintenance.

All check, adjustment, maintenance and lubrication steps must be carried out by staff previously trained and instructed.

All check, adjustment and maintenance steps of the electric system must be carried out by staff previously trained and instructed.

People on drugs, alcohol or drowsiness–causing medicines are not allowed to use or maintain the machine.

All check, adjustment, maintenance and lubrication steps must be carried out with the MACHINE STOPPED and DISCONNECTED POWER.

The cap/cork feeding system has been designed and manufactured to be loaded by an automatic elevator with ground loading.

If caps/corks are loaded manually, this operation must be performed with the machine off.

For interventions in the machine upper part, with MACHINE STOPPED and DISCONNECTED POWER, exclusively use climbing means (ladders, scaffolds) equipped with guards and according to the laws in force. Do not carry out any intervention climbing on the machine structure or parts.

16. SAFETY NORMS FOR THE MAINTENANCE

16.1. MAINTENANCE PREPARATION

BEFORE ALL MAINTENANCE OPERATIONS, the machine must be set in safety conditions to prevent danger of unexpected starts and/or electrocution.

- A sign "Maintenance in progress" must be placed on the control board.
- The section under maintenance must be fenced off to prevent unauthorized persons from accessing and must be signalled by plates: "Danger! Maintenance in progress".
- After the maintenance works and before starting the machine again it is always necessary:
  - to check that the possibly replaced parts and/or used tools have been removed from the machine;
  - to check that all guards, protections and safety devices possibly removed during the intervention have been re–installed correctly and are operating.

The standard operating conditions will be restored only after the intervention and the checks.

16.2. GENERAL SAFETY WARNINGS

The safety for the machine and the operators also depends on a regular maintenance according to the manufacturer's indications.

The routine and extraordinary maintenance interventions must be exclusively charged to skilled staff, that is to technicians qualified according to the specifications in the manual introduction.

If outsourced operators intervene, they must ensure the same performance of the inhouse qualified technicians and must work in safe operating conditions.
The qualified technicians must:

- respects the limits of their tasks (mechanical, electrical, etc.);
- follow the procedures and warnings in the manual, within their tasks;
- follow the times and intervals indicated in the manual for scheduled maintenance interventions.

The following general warnings must be noted:

- Electrocution hazard due to direct contact with the machine power supply terminal boards and in the electrical system connector blocks (the hazard is signalled by plates with the suitable triangle with yellow bottom).
- Be extremely careful during the interventions on the electrical system, if the system must be left live. Such interventions must be carried out by staff skilled in this type of maintenance only, trained and correctly equipped for the steps to be made. Interventions with the system live must always be previously authorized by the employer, who shall supervision them evaluating the specific risk conditions.
- Repair pneumatic systems and components only after having cut them off and having checked there is no residual pressure inside the components.
- It is forbidden to carry out maintenance and lubrication interventions on moving components.
- When working on heated parts, always wait for them to cool down to avoid burns or scalds.

16.3. WARNINGS FOR A CORRECT MAINTENANCE

For a good maintenance:

- follow the maintenance intervention intervals indicated in the manual; the interval (indicated in time or in working cycles) between interventions must be meant as max. limit not to be exceeded. If necessary, it can be shorter.
- A correct preventive maintenance requires a constant care and continuous monitoring of the system and of the machines. Promptly check the cause for possible faults such as excessive noise, overheating, etc. and solve it.
- In case of doubts, contact the manufacturer.

17. MACHINE CLEANING

It is recommended to use clothes wetted with preferably warm water and non–aggressive detergents for the machine general cleaning.

Wash the machine remaining below the capping area.

The customer shall refer to the instructions by the supplier of the detergents used for cleaning the machine.
It is absolutely forbidden to wash the machine with pressurized water jets or aggressive detergents.

18. SAFETY CHECKS

18.1. AT THE END OF THE MAINTENANCE INTERVENTIONS, BEFORE STARTING THE MACHINE UP

The qualified technician or staff performing the maintenance shall check that all the steps have been ended and all safety devices are correctly operating.

C

SCRAPPING AND DISPOSING

In this section are listed some important operations to carry out if the demolition of the machine is not carried out by specialized firm for the dismalting and scrapping, but takes place customer offices.

Proper precautions must be taken for out–of–order and scrapping of the machine in order to avoid possible dangerous situations.

In particular, follow these general instructions:

1) disconnect the machine from the electric network and check that no residual voltage exists;
2) empty all the lubrication points of oil paying attention in not dispersing and dispose it according the laws in forces in the country where the machine has been used.
3) lock all the moving parts of the machine, making sure that there are no possibilities of movement, also due to accidental collision during the transfer operation;
4) with the help of qualified and authorized personnel proceed to the dismounting of components and the differentiated collection of materials (steel, plastic, electrical components, wires, etc.) in order to dispose everything according to the indications forseen by the laws in force

Always use security clothes during the operations that require direct intervention on the machine

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D

UNLOAD AND INSTALLATION OF THE TURRET

1. UNLOAD FROM THE MEAN OF TRANSPORT

Mounting and connection operations of the turret to other units or components must be carried out by specialised personnel only.

Unload the baseframe complete with the turret as follows:

- hook using belts/chains of adequate capacity to hoist the weight indicated at point "**a.**" page 1, to eyebolts placed on the transfer baseframe
- hook the belts/chains to a crane and lift the turret.

Lifting the machine, make sure that the belts/chains do not excessively press parts of the turret.

Transport the machine near the bottling line.

Do not stand or walk under the machine when it is hoisted!

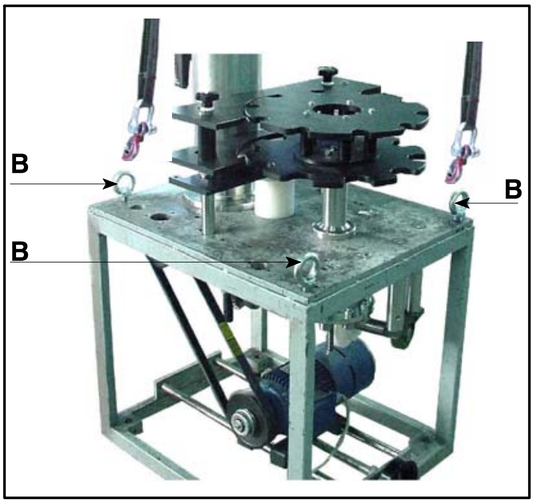

Fig.5

2. LAYING THE TURRET

2.1. REMOVAL OF THE COMPONENTS OF THE TURRET FROM THE TRANSFER BASEFRAME

- Unscrew bolts that fixing the rear orientable support (**U**) to the supporting column (**S**)

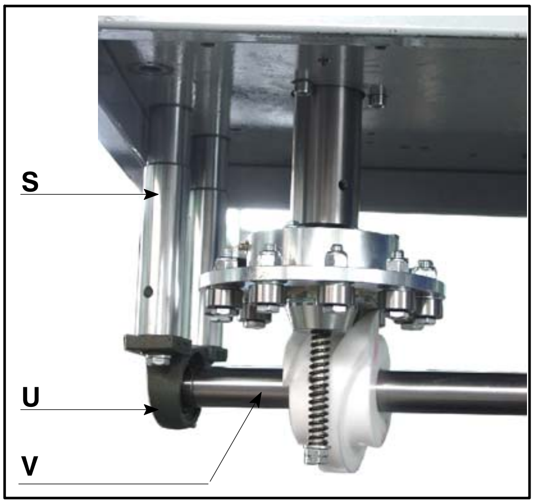

Fig.6

UNLOAD FROM THE MEAN OF TRANSPORT

1.

- Unscrew bolts that fixing the front orientable support (**Y**) to the flange (**T**)
- Remove the horizontal shaft (**V**)
- Unscrew bolts (**F**) fixing the reducer (**C**) to the fixed bush

Unscrewing bolts (F) adequately hold the complete reducer block

- Remove the reducer (**C**) complete with the vertical shaft
- Hook a lifting belt or rope to the fitting eyebolt (**A**) and tension it.

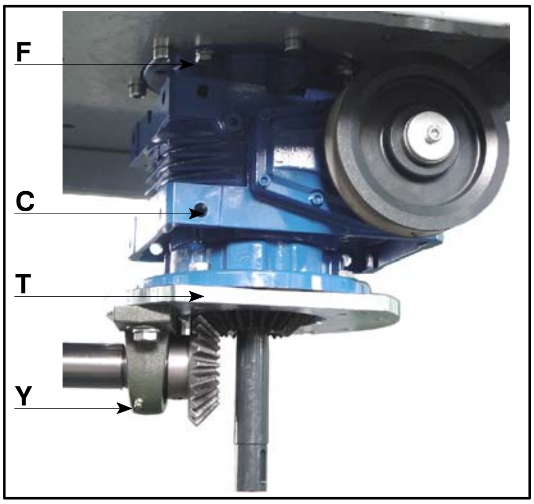

Fig.7

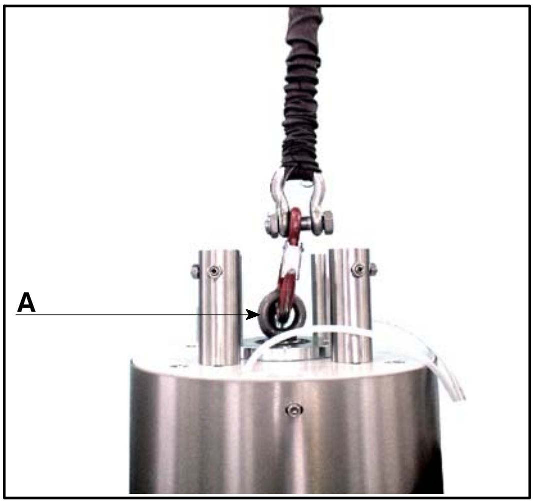

Fig.8

- From the lower part of the supporting baseframe plate unscrew the screws (**B**) and remove the upper part of the turret

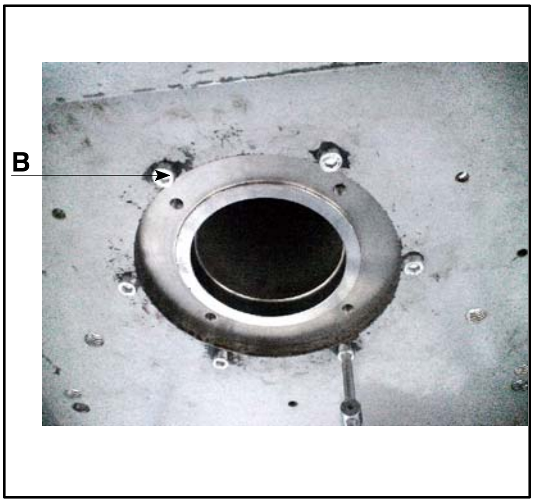

Fig.9

- Unscrew the locking nut and the nut (**G**) and unthread the washer and the clutch spring (**H**)

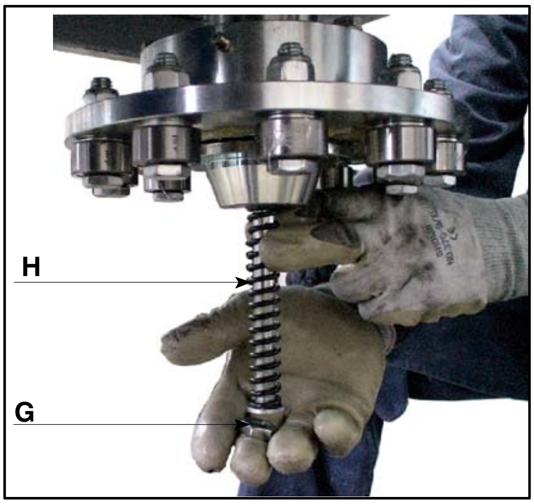

Fig.10

- Slide the balls supporting bush out (**I**)

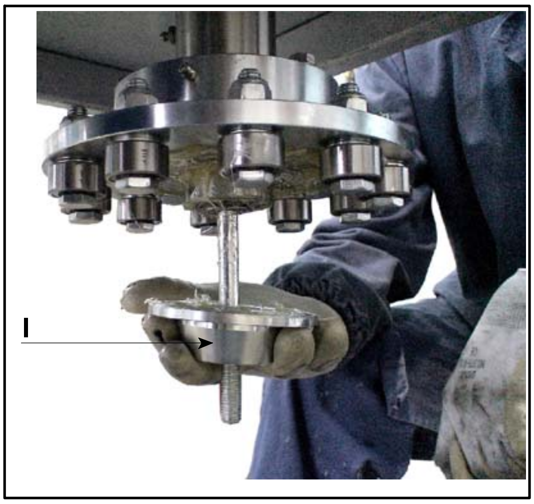

Fig.11

- Loosen the fixing grub screw (**L**)

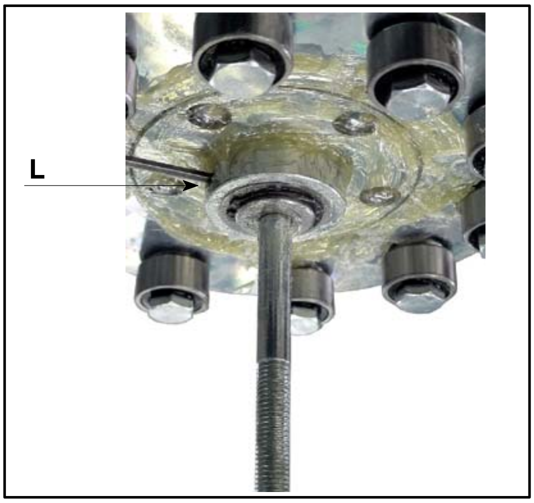

Fig.12

- Using a nose pliers remove the seeger (**M**) and unthread the bearings holder bush (**N**)

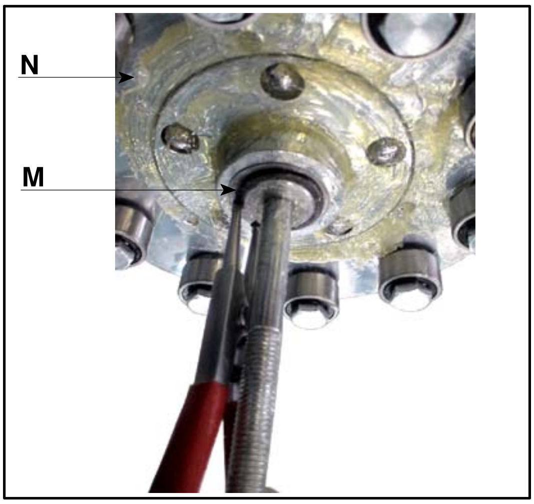

Fig.13

- Unscrew and remove the greaser (**O**)
- Remove the star–wheel support (**P**) from the fitting seat of the baseframe
- Remove the screw (**Q**) and unthread the star–wheel shaft from the baseframe plate

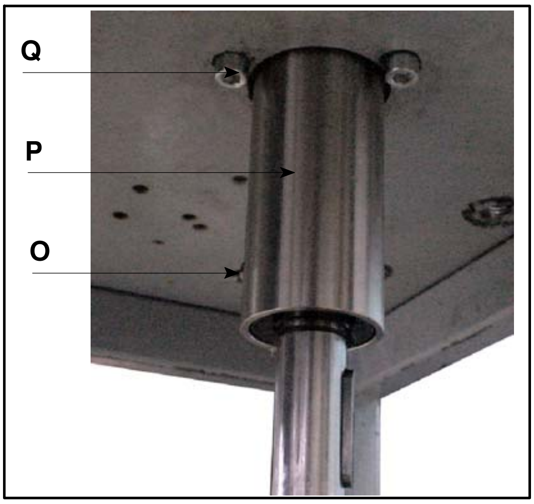

Fig.14

- Remove screws (**CB**) fixing the passage plane (**J**) to the passage plane support
- Unscrew nuts (**CC**) and remove them with the washers in such way to slide the passage plane out (**J**) from the supporting columns (**K**)
- Unscrew columns (**K**) of the baseframe plate

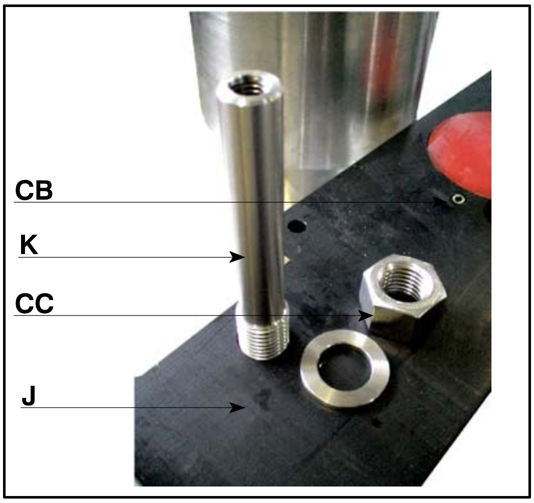

Fig.15

- Unscrew screws and remove the passage plane support (**W**) from the baseframe plate

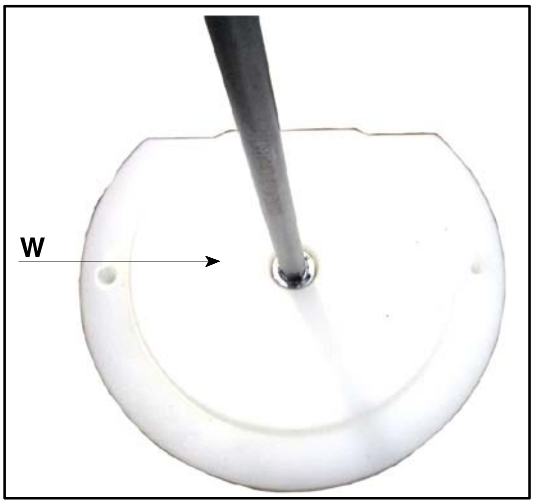

Fig.16

2.2. ASSEMBLING OF THE TURRET ON THE MONO-BLOCK

Make sure that the work place and the area for the installation of the turret are free.

The ground must be dry and without oil in such way to avoid slipping and falling during the transport, installation, connection, adjustment ,ecc.

- Hook a lifting belt or rope to the fitting eyebolt (**A**).

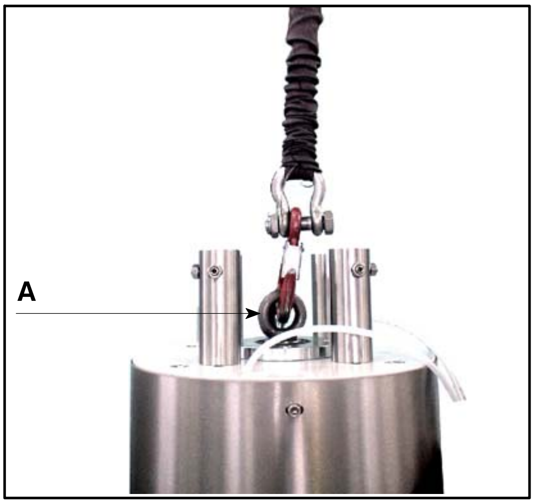

Fig.17

Use a rope or belt adequate for the turret lifting and in compliance to the safety rules in force, checking them before to use them to see if there are some defects.

- Lift the turret and place it into the fitting seat on the mono-block.

- Fix the upper part of the turret to the baseframe plate by screws (**B**). Screw from the lower part of the plate

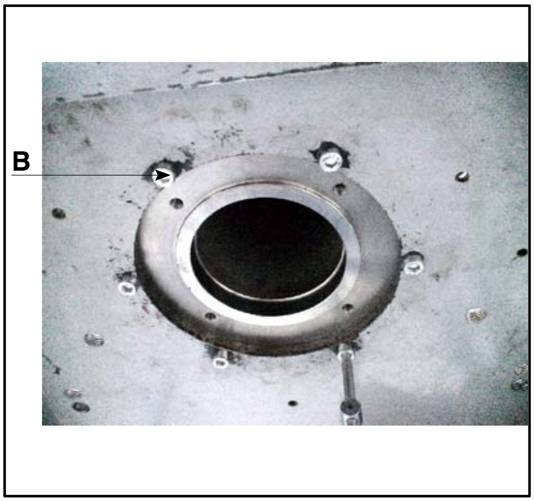

Fig.18

- Unhook the lifting belt and remove the eyebolt (**A**)
- Place the reducer (**C**) complete with shaft (**D**) and conical pin (**E**) under the plate of the monoblock.

*Check that the timing of the movement trasmission shaft (**D**) with the grooved shaft inside the fixed bush of the turret; references must be aligned.*

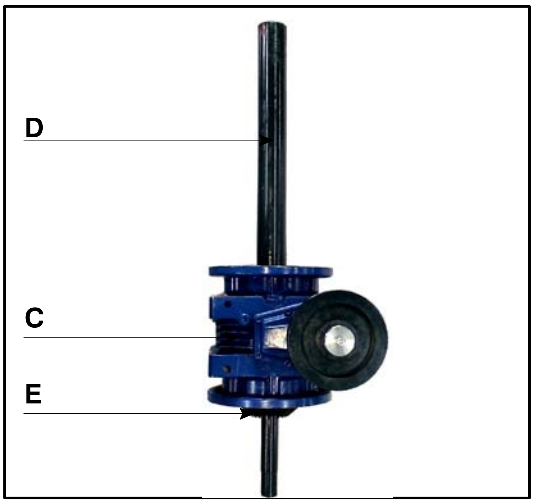

Fig.19

- Fix the reducer (**C**) to the fixed bush of the turret by bolts (**F**)

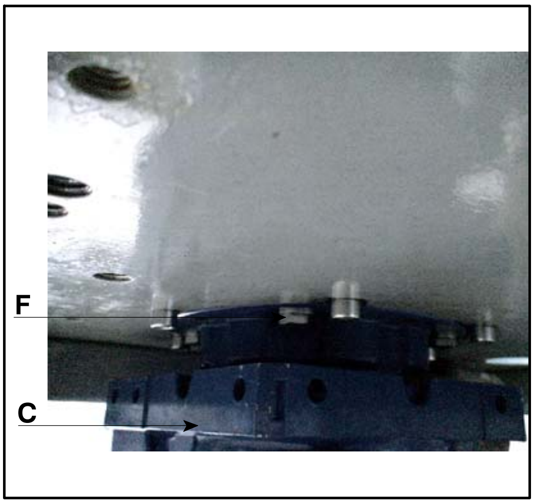

Fig.20

- Place the star–wheel support (**P**) into the fitting plate of the base-frame
- Fix the star–wheel shaft to the baseframe plate by screws (**Q**). Screw under the plate

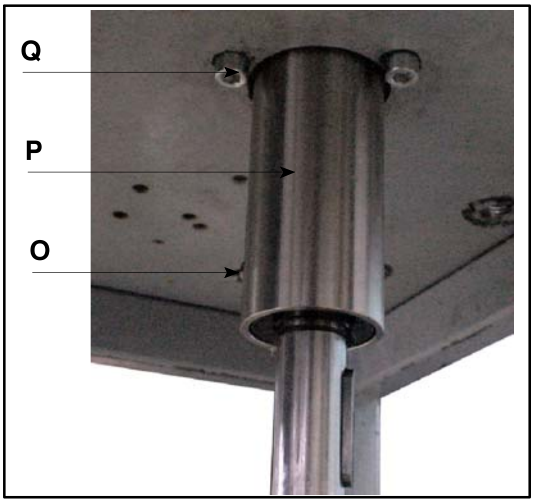

Fig.21

- Remount components on the star–wheel shaft proceeding in the reverse way to what described above
- Adjust the clutch torque carrying out on the nut and locking nut (**G**)

<u>For a correct adjustment, the lower part of the locking nut must be aligned with the lower part of the threaded rod of the star–wheel shaft</u>

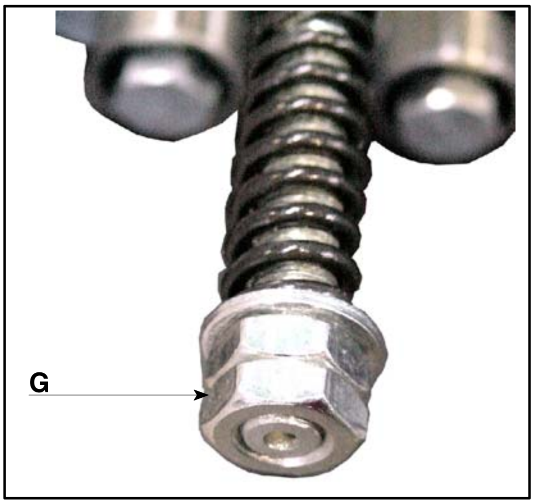

Fig.22

- Screw the microswitch holder rod (**R**) under the baseframe plate
- Adjust the height of the microswitch carrying out on the fitting clamp in such way that the actioner is in contact with the lower part of the balls supporting bush

Fig.23

- Fix the supporting column (**S**) under the baseframe plate

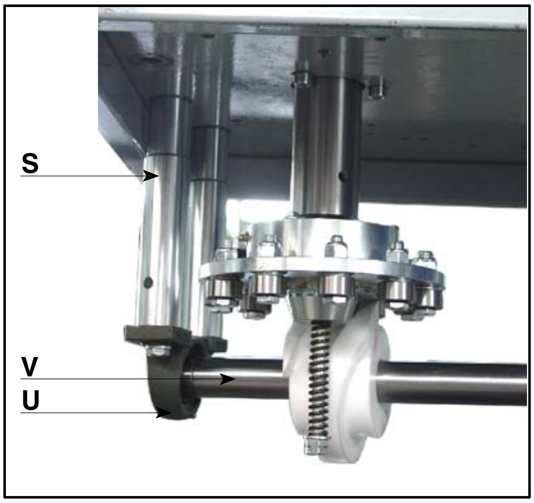

Fig.24

- Fix the flange (**T**) under the reducer (**C**)
- Fix the rear orientable support (**U**) complete with horizontal shaft (**V**), to the supporting column (**S**)
- Fix the front orientable support (**Y**) complete with conical pin (**Z**) to the flange (**T**)

*Check that the correct timing between the conical pin teeth (**E**) and (**Z**); the references indicated on pins must be aligned*

- Fix the passage plane support (**W**) on the baseframe plate

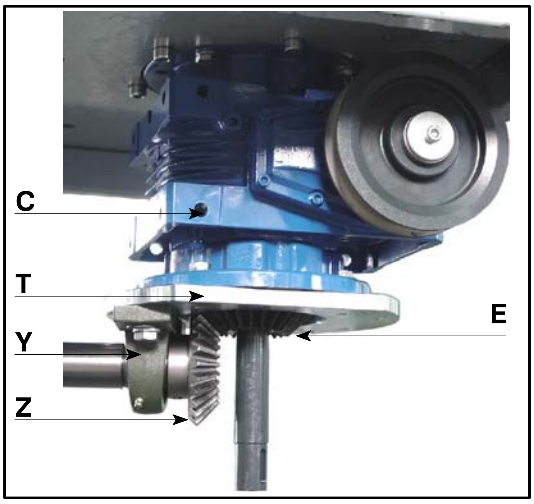

Fig.25

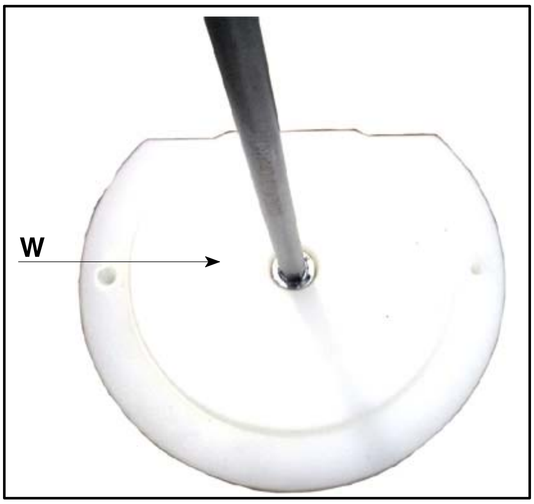

Fig.26

- Screw on the baseframe plate the columns (**K**)
- Place the bottles supporting plate (**X**) into the fitting seat of the passage plane (**J**)
- Install the passage plane (**J**) on the supporting column and fix it by the fitting washers and nuts
- Fix the passage plane (**J**) to the passage plane support (**W**)

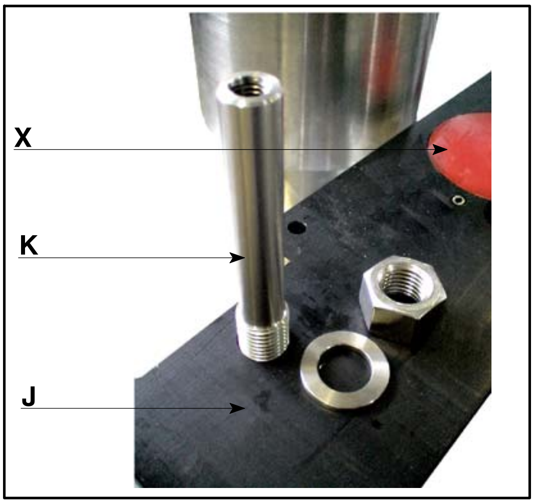

Fig.27

3. ELECTRIC CONNECTION

Interventions on the electric system can only be carried out by SPECIALISED ELECTRICIANS, trained about the electric characteristics of the machine and about safety regulations too.

FEEDING ELECTRIC PLANT must be built in accordance with law in force

The technical data relevant to the electric plant is shown in point "**d.**", page 2.

Connect the electric cables to the terminal stripes.

Both cable and relevant clamp to which must be connected have the same number.

Periodically check that the electric wirings are in perfect condition.

3.1. ROTATION DIRECTION CHECK

After the electric connections, check the that the rotation direction of motors of the turret (when provided) is the same of the one indicated by the arrow placed on the motors. The check can be carried out observing the rotation of cooling blades of the motor in object.

If the rotation is not correct, it is necessary to:

- invert the 2 phase power cables of the motor.
- repeat the same operations to check if the rotation is correct.

Check the correct plug–socket connections before machine power on.

4. COMPONENTS INSTALLATION

For the installation of the components of the machine, see the relevant chapters about these components. See the index for the consultation.

5. PNEUMATIC CONNECTION

Before carry out any operation make sure that the installation is not under pressure, and disconnect the feeding if necessary.

Carry out the connection to the network as follows:

- connect the main pneumatic unit coupling (**A**) to the distribution net,
- if necessary, open the sliding valve (**B**),

For pneumatic data see point “**e.**” page 2.

The pneumatic unit scheme is provided with the spare–parts catalogue.

Fig.28

E FIRST START UP, OPERATING AND USE

1. SETTING UP, CHECKS AND TEST FOR THE FIRST START UP

The stting up operations of the machine for the first start up must be carried out by qualified technicians.

The machine has been tested by the Manufacturer premises before the shipping; all the calibration and eventual adjustments have been carried out during the test.

1.1. CHECKS BEFORE THE START UP

In order to prevent errors and/or accidents, before the start up of the machine it is necessary to carry out some checks:

- check that all the electric feeding phases are correctly connected;
- check that the machine is correctly feeded, checking the tension;
- check that there are no leaks in the pneumatic unit ;
- check the correct motor absorption;
- check the correct rotation direction of motors.

2. OPERATION

The turret cannot work autonomously.

The customer must thus install the turret on his machine under his own responsibility and according to the European Community directives and take care that all requirements and specifications of the laws in force are complied with.

3. PRELIMINARY CHECKS

Before starting the turret up, check the following:

- check all safety devices are operating;
- check all machine components are lubricated (see chapter *LUBRICATION*);
- make sure the specific installed equipment are suitable for the size to be processed;
- check the correspondence between container and cork to be treated;

<u>Every set of equipment is marked as shown in the table ”**c.**” on page 1. Each mark corresponds to a size to be processed.</u>

- check the height of the closure unit according to the container height;
- after checking and possibly adjusting the turret height, make sure the equipment is aligned;

<u>To check the alignment, carry out a test making the machine perform a whole cycle at slow speed and in jog mode after having fed a few containers.</u>

- check the cork orientator is sufficiently full;
- check that the working pressure is correct (the pressure is indicated on the pressure gauges).

4. HEIGHT ADJUSTMENT OF THE TURRET

To carry out the adjustment and check, proceed as described below:

- unlock the screw (**A**) locking the column;
- turn the handwheel (**B**) in clockwise direction to lift the turret until to exceed the height of the bottle;
- lean a bottle of the format to handle on the plate (**C**);
- turn the handwheel (**B**) in counter clockwise direction to lower the turret until the spacer (**D**) is on the bottle;
- lock the screw (**A**);
- remove the bottle used for the adjustment;

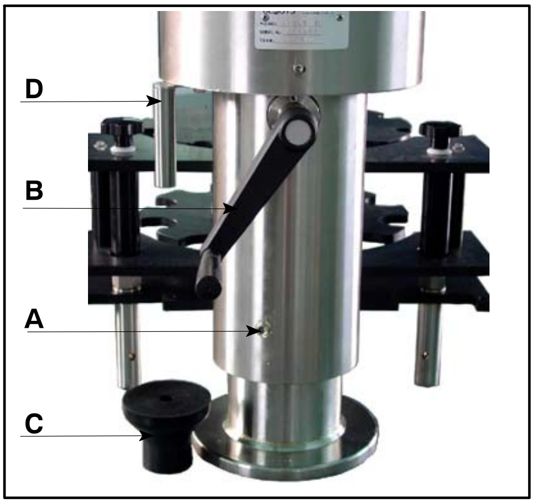

Fig.29

5. USE OF THE TURRET

5.1. HEAD ROTATION SPEED ADJUSTMENT

If it is necessary to adjust the rotation speed of the head it is necessary to carry out on the handwheel (**B**) of the variable speed−drive.

The adjustment of the head rotation speed must be carried out when the motor is in rotation.
The speed must be adjusted according to the production speed and the nuber of necessary rotations for the closure of the cap.

The capping head rotation speed is correctly adjusted when, at the detachement there is only one click.

If the head rotation is too high, there are other clicks before the exit of the bottle, due to rollers of the head that reprise the thread until, in some cases, to cut the cap, in this case decrease the rotation speed.

If there are no click, increase the speed in order to avoid that the cap is not correctly closed.

Carry out some closure tests before starting up the production.

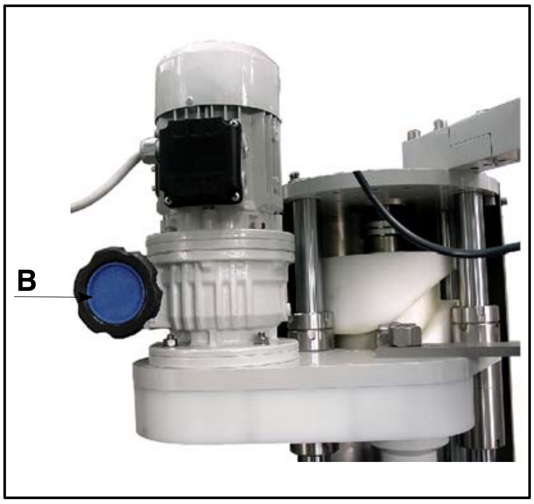

Fig.30

5.2. DISCONNECTION FROM THE WORK CYCLE

If it is necessary to exclude the turret from the work cycle, using the transfer belt for the container passage, it is necessary to:

- lift the turret so as to remove the bottle−guide equipment;
- remove the star−wheel,
- remove the conveyor,
- loosen bolts (**C**),
- turn the support of 180°,
- fix one of the bolts to lock the unit,

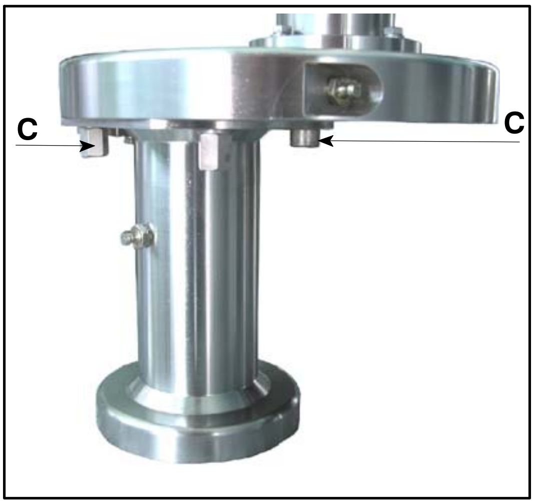

Fig.31

F

MAINTENANCE

1. PERIODICAL CHECKS AND MAINTENANCE

Routine checks and current maintenance described in this chapter aim at identifying operations necessary to keep the machine perfectly effective.

CLOSYS S.r.l. 's Technical Assistance is available for Customers in order to solve any problem concerning use and maintenance of machines

Requests must be forwarded after careful analysis of problems and causes, specifying the following:

- malfunctioning explanation in detail
- all checks carried out
- all adjustments carried out and their effects
- any other useful information

All the check, adjustement, maintenance and lubrication operations must be carried out by *qualified technicians*.

2. MAINTENANCE INTERVENTION

The following table has the maintenance time limit and the necessary operations for a correct operating and a long lasting during of the machine.

The described interventions are settled in intervals quantified in hours.

Interval (h)	*Working in 1 shift*	*Working in 2 shifts*	*Working in 3 shifts*
8	–	–	–
40	1 week	–	–
120	3 weeks	–	1 week
250	6 weeks	3 weeks	2 weeks
500	3 months	6 weeks	1 month
1000	6 months	3 months	2 months
3000	12 months	9 months	6 months
6000*	*	*	12 months
* every 6000 hours or anyway once a year.			

As an exemple in the table above there are correspondance calculated in weeks, months and years, considering daily workshifts of 8 hours each.

3. MAINTENANCE AND LUBRICATION PLANNING

operation	*intervention type*	*reference*
8 hours	lubrication	See chapter 5. on page 44
40 hours	routine maintenance	See chapter 4. on page 35
120 hours	lubrication	See chapter 5. on page 44
120 hours	routine maintenance	See chapter 4. on page 35
250 hours	lubrication	See chapter 5. on page 44
500 hours	lubrication	See chapter 5. on page 44
1000 hours	routine maintenance	See chapter 4. on page 35
12000 hours	routine maintenance	See chapter 4. on page 35
	extraordinary maintenance	See chapter 5. on page 39

4. PLANNED MAINTENANCE

Every 40 operating hours		
OPERATION	***PURPOSE***	***NOTES***
Air pressure check.	Check the air feeding pressure devices. Check that manometers of the machine mark the pressure indicated on the fitted plate, out on them.	
Cleaning of the threading head.	Clean the closure heads by a cleaning with lukewarm water and lubricate after the drying for the good operating of the seaming and in particulary the threading rollers.	
Check the heads roller.	Check the perfect rotation of the threading and seaming rollers (they must turn freely).	

<u>If compressed air is used (not recommended) it is necessary to wear protection gloves, earplugs and glasses, be careful to people close to the machine</u>

Every 120 operating hours		
OPERATION	***PURPOSE***	***NOTES***
Cleaning of the orientator.	Clean the orientator and check the correct feeding.	See the relevant chapter
Check the springs of the head.	Check the efficiency of the springs in general and if it is necessary replace them.	

<u>If compressed air is used (not recommended) it is necessary to wear protection gloves, earplugs and glasses, be careful to people close to the machine</u>

Every 1000 operating hours		
OPERATION	***PURPOSE***	***NOTES***
Check the threading head.	Check the state of wear of springs and if it is necessary replace them. Check the state of wear of rollers and mothpresser and if it is necessary replace them.	See the relevant chapter
Check the springs of the caps pick off gripper device.	Check the springs of the pick off gripper. If they are worn or damaged, replace them.	
Check of height sliding of the turret.	Check the correct sliding in height, letting the turret make an excursion from the minimum value to the maximum value, in such way to guarantee a good sliding of parts during the time.	

Equipment timing check.	Check the timing of the whole equipment and if it is necessary time it again.	See "*TRANSFER EQUIPMENT TIMING*" page 39.
Electric connections check.	Check and clean carefully all of the connections and, if necessary, replace worn parts. If any water infiltrations are found, even small ones, find immediately the source of the problem and eliminate it.	
Noises check (visual and auditory) clearances and cuts.	Check starting and stopping several times the machine without bottles,in order to check any abnormal noise, vibration in motorization components (motors reducers, gears and bearings). Replace all worn components. Lubricate the new installed components Clean and lubricate the dismounted components If there are any anomalies it is necessary to contact the *TECHNICIAN SERVICE.*	
Pneumatic system check.	Check proper operations and any leaks (fittings, pipes,etc.). ● check the air pressure at machine entrance ● feeding the pneumatic installation with compressed air, check that there are no leaks, broken or not connected pipes ● check that manometers of the machine mark the pressure indicated on the fitted plate, put on them.	

Every 12000 operating hours		
OPERATION	***PURPOSE***	***NOTES***
Machine general check.	Check the regular operating of the machine, if there are any anomalies it is necessary to contact the *TECHNICIAN SERVICE.*	

5. MAINTENANCE–TIMING

5.1. TRANSFER EQUIPMENT TIMING

5.1.1. STAR–WHEEL TIMING

In case the star–wheel is not timed, it is necesary to:

- place the star–wheel, turning it manually on itself until the clutch device come back to its original seat (there is a click);

If, after these operations, the star–wheel is not in the correct position, it is necessary to:

- turn the machine by pulses until the bottle is in the closure point and the head descends until to touch the cap
- loosen the locking screw (**CW**),

In this way the star–wheel is free to turn and so is possible to time it in such way that the pocket is coaxial to the head;

- firmly lock the screw (**CW**);

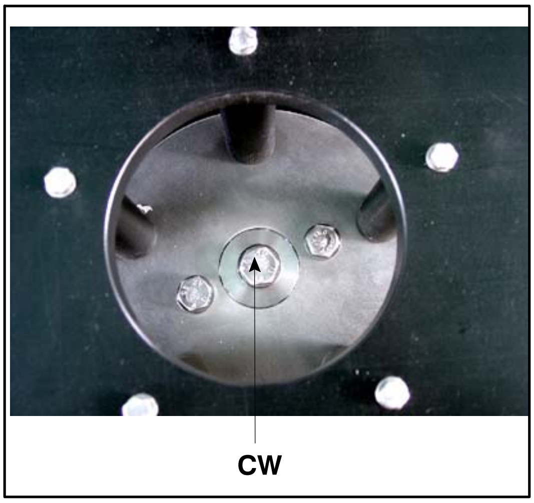

Fig.32.

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G

LUBRICATION

1. GENERAL INFORMATION

All the lubrication operations must be carried out when the machine is stopped.

Before carry out any lubrication operation turn the machine on *"Maintenance State"*.

Before to start up the machine, check the lubrication of all the parts described in this section.

If the machine is not used for a long time, carry out again the lubrication.

Mineral oils, in case the machine is not use or has been stoppen for more than six months, lose their characteristics and must be replaced.

2. LUBRICANTS HOMOLOGATION ACCORDING TO USDA

The homologation according to the USDA provides two cathgories: USDA – H1 and USDA – H2

- **H1** these initials mark Food grade lubricants which are lubricants that could be used in every friction points of food and pharmaceutical industry machine and equipements, in which contacts between lubricants and food considered technically inevitable.
- **H2** these initials mark lubricants recommended for general use in food and pharmaceutical industry, supposing that there are no contacts with food.

3. LUBRICANT TYPES

The *R&D* department of *CLOSYS S.r.l.,* after careful studies and functional tests, has decided to use the lubricants indicated in the table below to lubricate its machines:

		KLÜBER LUBRICATION	MOLYKOTE FROM DOW CORNING	PETRO-CANADA	DUPONT
A	grease	Klübersinth UH1 64–62	G4501	Purity FG grease	–
B	grease	Klübersynth UH1–14–222	G0052	Purity FG grease	–
C	oil	Paraliq P 150	L0115FG	Purity Chain Oil Light	–
D	oil	Klubersummit FG46	L0346FG	Purity AW 46	–
E	grease	–	–	PC Purity Synth	–
F	grease	Klüberpaste UH1 84–201	P1900	–	–
G	grease	Barrierta L55/1	–	–	Krytox GPL 226
H	grease	Klübersynth UH1 64–62	G4501	Purity FG grease	–
I	oil	–	Multi oil spray	Purity MF spray	–
J	grease	Klübersynth UH1 64–62	G4501	Purity FG grease	–
K	grease	Paraliq GB363	G0052	Purity FG grease	–
L	grease	Paraliq GB363	G0052	Purity FG grease	–
M	grease	Klübersynth UH1–14–222	G0052	Purity FG grease	–
N	oil	Klüberoil 4 UH1 220 N	L1122FG	Purity FG gear oil 220	–

3.1. FEATURES OF THE USED LUBRICANTS

Product name	NSF	NLGI	Basic oil	Thickener	Vis-cosity index	Basic oil vis-cosity (cSt)	Processed penetration (0.1 mm)	Min. temp. (˚C)	Max. temp. (˚C)
KLUBER									
Klübersinth UH1 64–62	H1	2	PAO	gel–silica		60	265–295	–40	150
Paraliq P 150	H1		WO			150		–15	100
Klubersummit FG46	H1		PAO		130	46		–45	135
Klüberpaste UH1 84–201	H1	1	PAO	PTFE		200			
Barrierta L55/1	H1	1	PFPE	PTFE		400		–40	260
Paraliq GB363	H1	2	MO	silicate			215–245	–30	140
Klübersynth UH1–14–222	H1	2	PAO	complex aluminum		220	265–295	–35	100
Klüberoil 4 UH1 220 N	H1		PAO		150	220		–25	120
MOLYKOTE									
G4501	H1	1	PAO	complex aluminum		100	310–340	–40	150
G0052	H1	2	MO	complex aluminum		115	280	–12	150
L0115FG	H1		MO		100	140		–18	
L0346FG	H1		MO		100	46		–21	
P1900	H1	1	MO	complex aluminum		85	290–340	–30	300
Multi Oil Spray	H1		MO		102	96		–10	120
L1122FG	H1		PAO/ MO		142	197		–33	
PETRO–CANADA									
Purity FG grease	H1	1.5	HT/ MO	complex aluminum		160	330	–15	150
Purity AW 46	H1		HT/ MO		108	46		–15	
PC Purity Synth	H1	2	PAO	complex sulfonate calcium		46	294	–45	200
Purity MF spray	H1		HT/ MO		150	151		–9	150
Purity FG Gear Oil 220	H1		HT/ MO		92	206		–18	
Purity Chain Oil Light	H1		HT/ MO		150	151		–9	150
DUPONT									
Krytox GPL 226	H1	2	PFPE	PTFE	155	240		–30	288

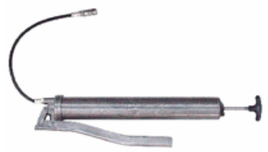
3.2. LUBRICANT RE–ORDERING

The lubricants used for the machine can be directly ordered to *CLOSYS S.r.l.* or to the local distributors.

See the table here below for the international addresses.

KLÜBER LUBRICATION	http://www.klueber.com
MOLYKOTE FROM DOW CORNING	http://www.molykote.com
PETRO-CANADA	http://www.petro–canada.ca
DUPONT	http://www.krytox.com

4. KEY TO THE LUBRICATION SCHEMES

SYMBOL	***DESCRIPTION***
	Lubricate with manual pump for grease
	Lubricate with manual pump for oil
	Lubricate with brush for grease
	Lubricate with spray bottle

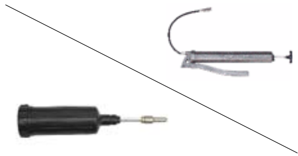
<u>Grams are referred to each lubrication point.</u>

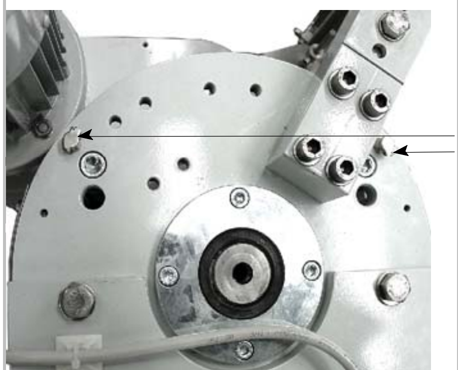
5. POINTS TO BE LUBRICATED

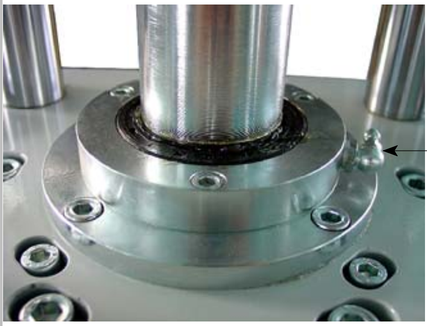
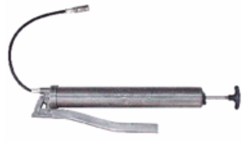
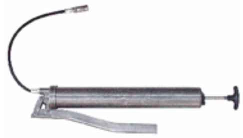
To protect the machine from wear, seizure or other severe damage to the mechanisms, it is necessary to lubricate and grease periodically all indicated points.

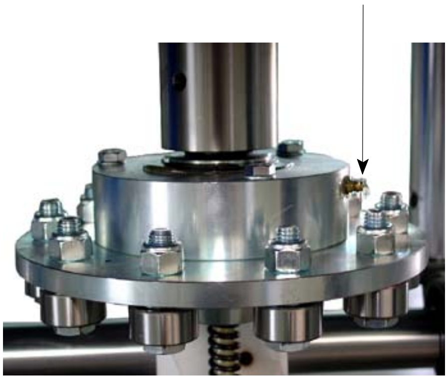
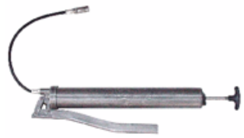
<u>*CLOSYS S.r.l.* guarantees the maintenance intervals indicated hereby and the correct operation of the mechanisms exclusively with these lubricants.</u>

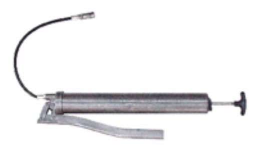
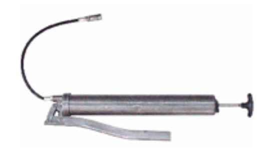
Every 8 hours lubrication		
LUBRICATION POINTS		***DESCRIPTION***
See the chapter about the head.		Lubricate: ■ Arms rollers. ■ Springs of seaming and threading rollers. ■ Seaming and threading rollers.
	I	

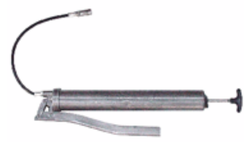
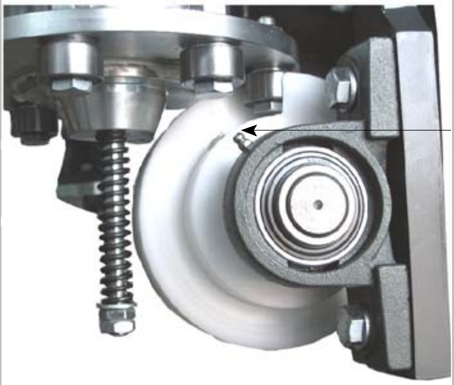
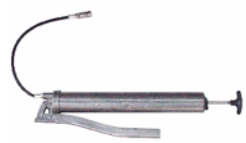
Every 120 hours lubrication		
LUBRICATION POINTS		***DESCRIPTION***
See the chapter about the head.		Lubricate the head central block
	K	

Every 250 hours lubrication		
LUBRICATION POINTS		***DESCRIPTION***
		Lubricate the sliding columns
	0,15 gr.	
	N	

	 **5 gr.** **A**	Lubricate the central bearings.
	 **5 gr.** **A**	Lubricate the sliding roller.

Every 500 hours lubrication		
LUBRICATION POINTS		***DESCRIPTION***
	 **5 gr.** **M**	Lubricate the star–wheel shaft

		10 gr. **M**	Lubricate the star–wheel shaft bearings.
		10 gr. **M**	Lubricate the star–wheel shaft gearings.
		**M**	Lubricate the motion trasmission gearing
		**M**	Lubricate the star–wheel rotation cam

	 **3 gr.** **M**	Lubricate the front orientable support
	 **3 gr.** **M**	Lubricate the rear orientable support
	 **A**	Remove the cover to lubricate. Lubricate the spindle controlling cam

6. LUBRICATION OF THE VARIABLE–SPEED DRIVE

6.1. INTRODUCTION

On the variable–speed drive there is an identification plate as the one that follows. On this plate there are all the references and all the indispensable indications for its correct identification.

Fig.33

Variable–speed drive V0.25 and V0.5

They are supplied with a load of synthetic lubricant oil “long life” Shell Donax TX type and have not a service plug. If there are not external contamination, lubricant periodical replacement usually are not carried out.

Variable–speed drive V1 ...... V5.5

It is recommended to carry out a first lubrication replacement after about 300 operating hours, and carefully clean the whole unit with suitable detergents.

According to the temperatures that are reached, the lubricant replacement will be carried out according to the intervals indicated in the table below:

Oil temperature t [˚C]	**Lubricant replacement interval [h]**
$65 \leq t \leq 80$	4000
$80 \leq t \leq 95$	2000

6.2. LUBRICATION

Before starting up the variable–speed drive, check the oil level through the glass, where provided. If it is necessary fill it or fill it to the brim.

Variable –speed drive V 0.25 and V 0.5 are supplied with a load of synthetic lubricant oil “long life” **Shell Donax TX** type and have not a service plug. If there are not external contamination, lubricant periodical replacement usually are not carried out. If it is necessary a filling to the brim, or the complete replacement of the lubricant, follow the indication and quantities specified in the following tables.

Do not mix oils with different nature. Eventual replacement, or filling to the brim, must be carried out with the same type of lubricant.. If the same oil is not disposible, completely empty the variable–speed drive, and clean it with a light solvent, before the next oil filling.

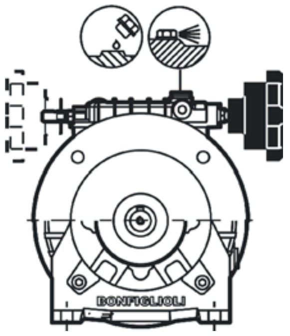
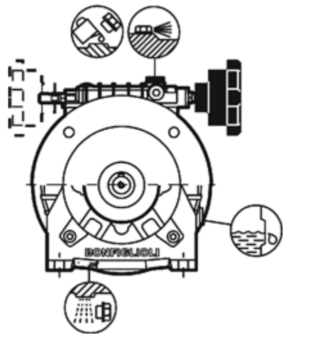
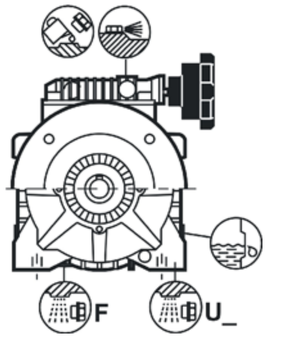
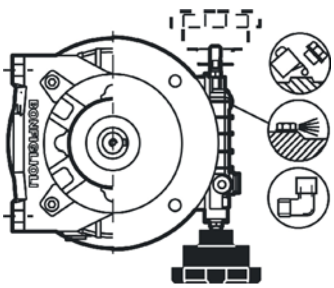
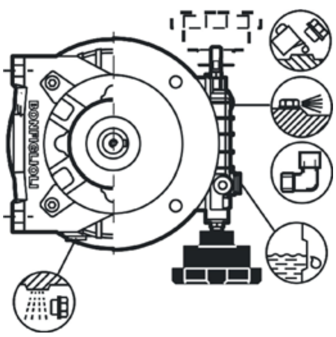
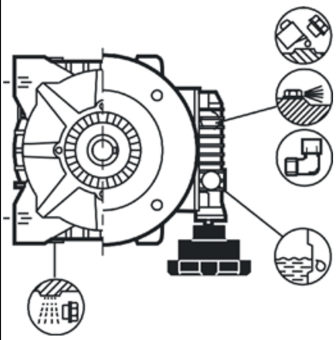
Shell Donax TX			
Volumetric mass	ISO 3675	0,852	kg/dm^3
Kinematic viscosity at 40° C	ISO 3104	34	cSt
Kinematic viscosity at 100° C	ISO 3104	7,4	cSt
Viscosity index	ISO 2909	196	–
Flame point	ISO 2592	198	°C
Sliding point	ISO 3016	–48	°C

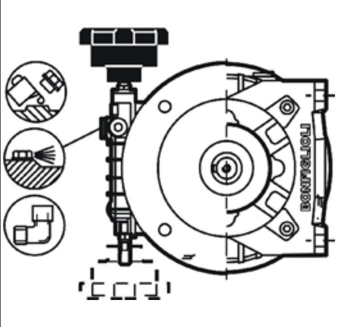
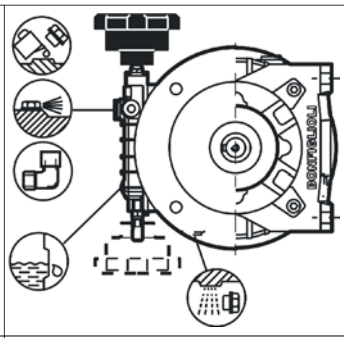
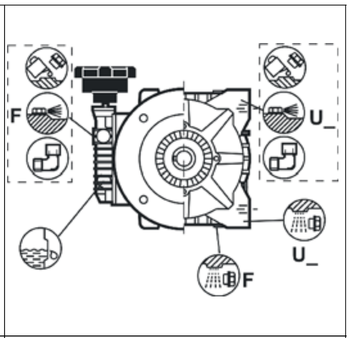
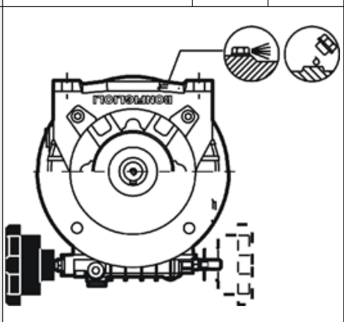
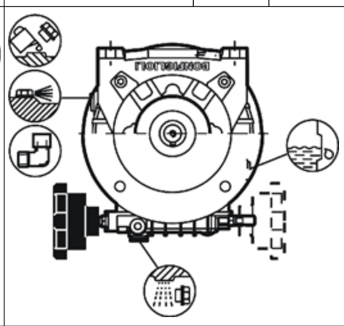
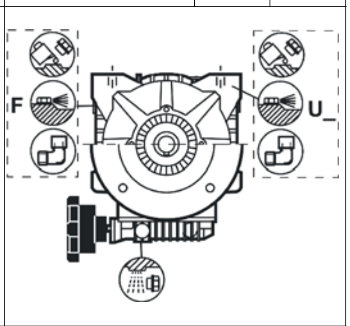
Variable–speed drive from V1 to V5.5 are supplied with a load of mineral lubricant oil **Shell Donax TA** type**.**

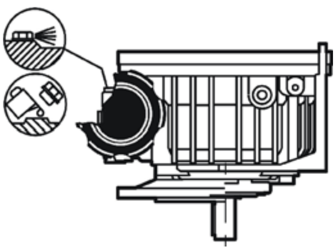
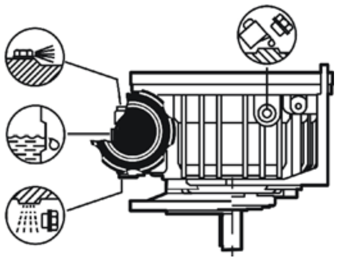
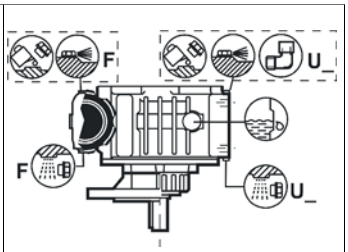
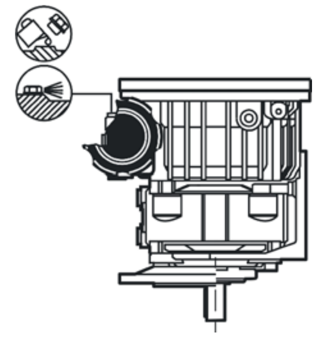
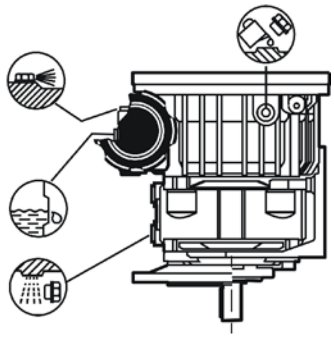
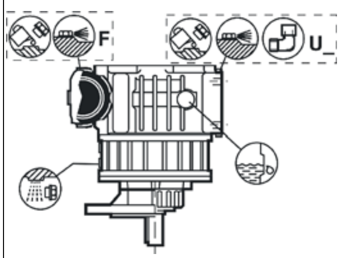
Eventual replacements or fillings to the brim can be carried out using compatible lubricants.

Anyway, do not mix mineral and synthetic oils.

Shell Donax TA			
Volumetric mass	ISO 3675	0,873	kg/dm^3
Kinematic viscosity at 40° C	ISO 3104	37,3	cSt
Kinematic viscosity at 100° C	ISO 3104	7,0	cSt
Viscosity index	ISO 2909	151	–
Flame point	ISO 2592	196	°C
Sliding point	ISO 3016	–42	°C

	V 0.25 – V 0.5			V 1 – V 2			V 3 – V 5.5		
B3							 		
	Donax oil TX (for life)			Donax oil TA (f2000–3000 hours)			Donax oil TA (2000–3000 hours)		
	V 0.25 F V 0.5 F	0.14 0.18		V 1 F V 2 F	0.30 0.40		V 3 F/V 5.5 F	0.70	
	V 0.25 U... VR 0.25...	0.12		V 1 U... VR 1...	0.25		V 3 U VR 3	1.00	
	V 0.5 U... VR 0.5...	0.15	oil	V 2 U... VR 2...	0.32	oil	V 5.5 U VR 5.5	1.00	oil
	VD 0.5 U... VRD 0.5...	0.30	[l]	VD 1 U... VRD 1...	0.35	[l]	VD 3 F VD 5.5 F	1.30	[l]
	–	–		VD 2 U... VRD 2...	0.46		VD 3 U VRD 3 U	1.60	
	–	–		–	–		VD 5.5 U VRD 5.5 U	1.60	
B6									
	Donax oil TX (for life)			Donax oil TA (f2000–3000 hours)			Donax oil TA (2000–3000 hours)		
	V 0.25 F V 0.5 F	0.14 0.18		V 1 F V 2 F	0.30 0.40		V 3 F/V 5.5 F	0.90	
	V 0.25 U... VR 0.25...	0.12		V 1 U... VR 1...	0.25		V 3 U VR 3	1.00	
	V 0.5 U... VR 0.5...	0.15	oil	V 2 U... VR 2...	0.32	oil	V 5.5 U VR 5.5	1.00	oil
	VD 0.5 U... VRD 0.5...	0.30	[l]	VD 1 U... VRD 1...	0.35	[l]	VD 3 F VD 5.5 F	1.30	[l]
	–	–		VD 2 U... VRD 2...	0.46		VD 3 U VRD 3 U	1.60	
	–	–		–	–		VD 5.5 U VRD 5.5 U	1.60	

B7									
	Donax oil TX (for life)			Donax oil TA (2000–3000 hours)			Donax oil TA (2000–3000 hours)		
	V 0.25 F V 0.5 F	0.14 0.18	oil [I]	V 1 F V 2 F	0.30 0.40	oil [I]	V 3 F/V 5.5 F	0.90	oil [I]
	V 0.25 U... VR 0.25...	0.12		V 1 U... VR 1...	0.25		V 3 U VR 3	1.00	
	V 0.5 U... VR 0.5...	0.15		V 2 U... VR 2...	0.32		V 5.5 U VR 5.5	1.00	
	VD 0.5 U... VRD 0.5...	0.30		VD 1 U... VRD 1...	0.35		VD 3 F VD 5.5 F	1.30	
	–	–		VD 2 U... VRD 2...	0.46		VD 3 U VRD 3 U	1.60	
	–	–		–	–		VD 5.5 U VRD 5.5 U	1.60	
B8									
	Donax oil TX (for life)			Donax oil TA (f2000–3000 hours)			Donax oil TA (2000–3000 hours)		
	V 0.25 F V 0.5 F	0.14 0.18	oil [I]	V 1 F V 2 F	0.30 0.40	oil [I]	V 3 F/V 5.5 F	1.00	oil [I]
	V 0.25 U... VR 0.25...	0.12		V 1 U... VR 1...	0.25		V 3 U VR 3	1.30	
	V 0.5 U... VR 0.5...	0.15		V 2 U... VR 2...	0.32		V 5.5 U VR 5.5	1.30	
	VD 0.5 U... VRD 0.5...	0.30		VD 1 U... VRD 1...	0.35		VD 3 F VD 5.5 F	1.60	
	–	–		VD 2 U... VRD 2...	0.46		VD 3 U VRD 3 U	1.90	
	–	–		–	–		VD 5.5 U VRD 5.5 U	1.90	

V5									
	Donax oil TX (for life)			Donax oil TA (2000–3000 hours)			Donax oil TA (2000–3000 hours)		
	V 0.25 F V 0.5 F	0.28 0.30	oil [l]	V 1 F V 2 F	0.58 0.78	oil [l]	V 3 F/V 5.5 F	2.10	oil [l]
	V 0.25 U... VR 0.25...	0.22		V 1 U... VR 1...	0.40		V 3 U / VR 3	2.00	
	V 0.5 U... VR 0.5...	0.27		V 2 U... VR 2...	0.54		V 5.5 U VR 5.5	2.00	
V5									
	Donax oil TX (for life)			Donax oil TA (for life)			Donax oil TA (2000–3000 hours)		
	VD 0.5 U... VRD 0.5...	0.70	oil [l]	VD 1 U... VRD 1...	1.00	oil [l]	VD 3 F VD 5.5 F	4.50	oil [l]
	–	–		VD 2 U... VRD 2...	1.50		VD 3 U VRD 3 U	4.80	
	–	–		–	–		VD 5.5 U VRD 5.5 U	4.80	

6.3. OIL REPLACEMENT FOR V1.....V5.5

1) Place a container of adecuate capacity under the discharge plug.
2) Remove the charge, discharge and level plugs and let the oil freely flow.

To make easier the discharge operation, it is recommended to operate with hot oil.

3) Wait some minutes so that all the oil is went out, then screw again the discharge plug, after the gasket has been replaced.
4) Put the variable–speed drive in its definitive position and pour oil. Pour slowly the lubricant to make easier an uniform filling. Stop when the level hole is reached. Screw again the level plug, afte the gasket has been rerplaced, and complete the filling until the half has been reached.
5) Screw the charge plug, after the gasket has been replaced.

The exhausted oil must be disposed according to the environment protection norms

If there are leaks, before to reset the lubricant quantity, it is necessary to find the cause of the problem before startin up again the variable–speed drive.

7. REDUCER LUBRICATION

7.1. INTRODUCTION

On the reducer there is an identification plate as the one that follows. On this plate there are all the references and all the indispensable indications for its correct identification.

Fig.34

7.2. LUBRICATION

Reducers VF30....VF49 and W63....W86

For reducers of this group, that are on a average at low power, the permanent lubrication with synthetic oil is adopted. These reducers have not charge, discharge and level oil plugs and so do not need any maintenance having already the right oil quantity.

Reducers VF130....VF150 and W110

For reducers of this group, that are on a average at high power, the oil lubrication is adopted. Reducers have charge, discharge and level oil plugs

Oil quantity table (litres)

		B3	B6	B7	B8	V5
VF30		0.045	0.045	0.045	0.045	0.045
VF44		0.075	0.075	0.075	0.075	0.075
VFR44		0.050	0.050	0.050	0.050	0.050
VF49		0.12	0.12	0.12	0.12	0.12
VFR49		0.065	0.065	0.065	0.065	0.065
VF130	HS	3.9	2.5	2.5	2.3	3.3
	P(IEC)	3.0	2.5	2.5	2.3	3.3
VFR130		0.40	0.50	0.50	0.70	0.40
W63	i=7...15	0.31	0.31	0.31	0.31	0.31
	i=19...100	0.38	0.38	0.38	0.38	0.38
W75	i=7...15	0.48	0.48	0.48	0.48	0.48
	i=30,40	0.52	0.52	0.52	0.52	0.52
	i=20...100	0.56	0.56	0.56	0.56	0.56
W86	i=7...15	0.64	0.64	0.64	0.64	0.64
	i=30	0.73	0.73	0.73	0.73	0.73
	i=20...100	0.90	0.90	0.90	0.90	0.90
W110	P80...P132	1.5	1.7	1.7	1.9	1.7
	M2–M3	1.5	1.7	1.7	1.9	1.7
	$7 \leq i \leq 15$	1.5	1.7	1.7	1.9	1.7
	$20 \leq i \leq 100$	2.7	1.7	1.7	1.9	1.7

Reducers are supplied with a lubrication oil load **Shell Tivela oil S320** type.

It is recommended, if the lubricants is not of the recommended ones, that it has an equivalent composition on synthetic nature and viscosity. Moreover it must have the suitable antifoam additives.

7.3. OIL REPLACEMENT FOR VF130....VF150 AND W110

Do not mix oil with different trade marks and characteristics and check that the oil in use has high anti–foam characteristics and EP.

If an identic lubricat is not available completely empty the reducer and clean it inside with a light solvent, before the next filling.

1) Place a container of adecuate capacity under the discharge plug.
2) Remove the charge, discharge and level plug and let the oil freely flow.

To make easier the discharge operation, it is recommended to operate with hot oil.

3) Wait some minutes so that all the oil is went out, then screw again the discharge plug, after the gasket has been replaced.
4) Pour new oil only afterr the reducer is its definitive position, until the half of the level plug is reached
5) Screew the charge plug, after the gasket has been replaced.

The exhausted oil must be disposed according to the environment protection norms

If there are leaks, before to reset the lubricant quantity, it is necessary to find the cause of the problem before startin up again the reducer.

H

FORMAT CHANGE

1. PREMISE

All the equipment change operations must be carried out when the machine is not operating.

Close pneumatic feeding.

<u>The machine has been designed to handle containers and corks described in table “**c.**” page 1. It is necessary that each container and cap is handled exclusively with the equipment indicated in the table</u>

Each series of equipment is marked according to the format to handle:

- container body guide equipments are marked by coloured wedges.
- caps guide equipments, in case of several formats, are marked by an identification letter.

See table “**c.**” page 1 for the formats identification.

<u>When delivered the machine is ready to apply one of the types of containers and caps for which has been manufactured.</u>

To carry out correctly and without obstructions the format change operations it is necessary to lift the turret to its maximum height as described in paragraph 4. page 31.

2. CONTAINERS FORMAT CHANGE

If it is necessary to change the format of the container to handle it could be necessary to proceed as described in section “*INSTALLATION AND REMOVAL OF THE BOTTLES TRANSFER EQUIPMENT*”:

- replace the conveyor,
- replace the star–wheel,

Moreover it is necessary to adjust in height the turret (see par. 4. page 31).

3. CAP FORMAT CHANGE

When it is necessary to change the type of cap to apply, it could be necessary to proceed as described in section “*COMPONENTS ON THE MACHINE*”, replace the:

- caps distribution unit,
- orientating unit,
- closure head

CLOSYS

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I INSTALLATION AND REMOVAL OF THE BOTTLES TRANSFER EQUIPMENT

1. CONVEYOR

1.1. REMOVAL

- Unscrew fixing knobs (**CN**) of the conveyor.
- Remove the conveyor (**CO**).

1.2. INSTALLATION

- Place the conveyor (**CO**) into fitting seats.
- Fix by knobs (**CN**).
- Adjust the position of the conveyor in the following way:
  - insert two bottles into the pockets of the star–wheel
  - near the conveyor to the bottles.

<u>check that bottles have a minimum clearance in such way to allow a regular flow</u>

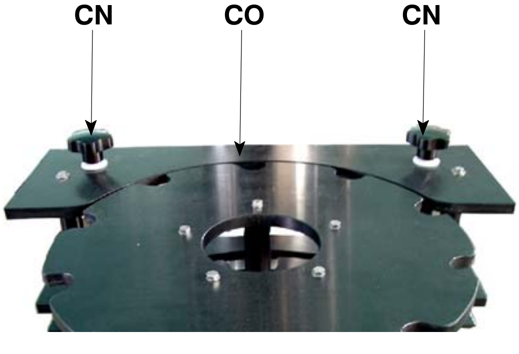

Fig.35

2. STAR–WHEEL

2.1. REMOVAL

- Unscrew bolts (**CR**).
- Remove the star–wheel (**CS**).

2.2. INSTALLATION

- Place the star–wheel into the fitting seats.
- Lock by bolts (**CR**).

Fig.36

Fig.37

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J

COMPONENTS ON THE MACHINE

In the following chapters, there are, in a complete way, the instructions about the components that are supplied with the machine.

The instructions about the components are grouped in only one chapter for each component, in such way the customer can easly refere the information about thair use.

The operations indicated in this section are for the *qualified technician.*

1. THREADING HEAD

All operations on the machine must be made when it is stopped.

1.1. TOOLS NEEDED FOR DISMOUNTING AND MAINTENANCE

No.		Wrench	Supplied
6		Wrench for ring nut 58–62–65	YES
7		Wrench for ring nut 34–36	YES
8		Hexagonal wrench.	YES
9		Hexagonal wrench 19.	YES
10		Hexagonal wrench 10.	YES
11		Wrench for socket head screws 3 mm	YES
12		Tube hexagonal wrench 6–7	YES
13		Noose wrench	YES
14		Adjustment caliper	YES

1.2. THREADING HEAD REPLACEMENT/REMOVAL

To replace or remove the threading head carry out the following operations:

1.2.1. REMOVAL

- insert the wrench (**6**) into seats (**A**) of the threading head (**B**) and unscrew to remove the head; if it is locked, it is sufficient to give a light stroke on the handle of the wrench;
- complete the head removal, loosening by hand.

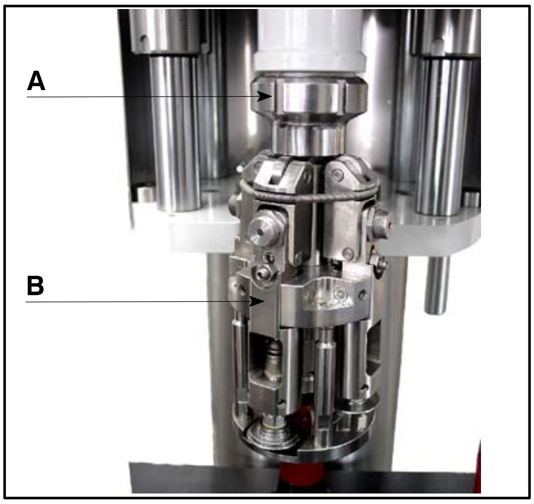


Fig.38

1.2.2. MOUNTING

- Center the seat of the key (**C**) and put manually the thread on the head of the spindle;

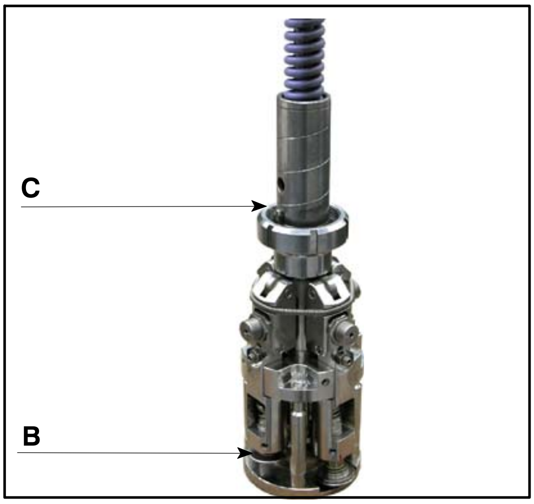


Fig.39

- complete the screwing of the head using the wrench (**6**).

1.3. HEAD ADJUSTMENT

The head is already delivered for the type of cap ordered by the customer and does not need any other adjustment. If it is necessary to proceed to a new adjustment (after the periodical maintenance or for a format change) follow the instructions described here below.

Remove the head from the machine as described in paragraph 1.2. on page 63.

1.3.1. HEAD PREPARATION

- Insert the head in a jaw with soft claws;
- using the wrench (**11**), unscrew the fixing Allen screws (**E**) and remove the centering device (**F**);
- using the wrench (**10**), unscrew and remove the studs (**G**) of the centering device.
- insert the suitable adjustment caliper (**14**), indicated in the adjustment table, on the mouthpresser;
- take the holder arms to the closing position pushing the head downwards;

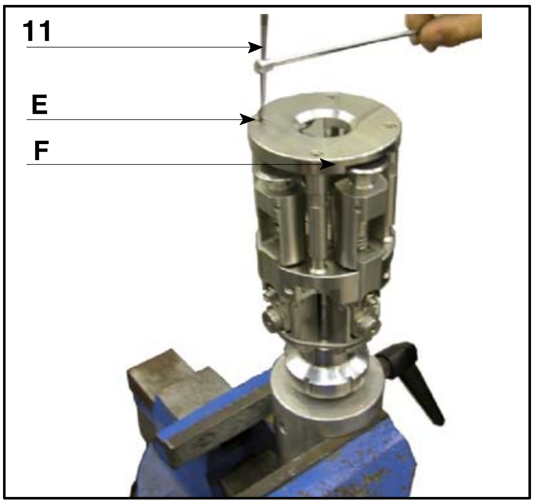


Fig.40

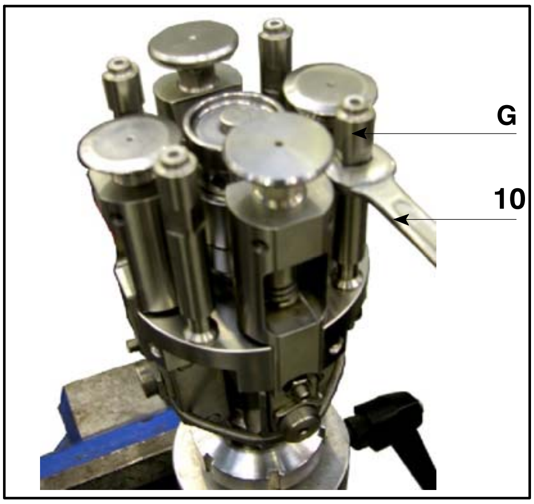


Fig.41

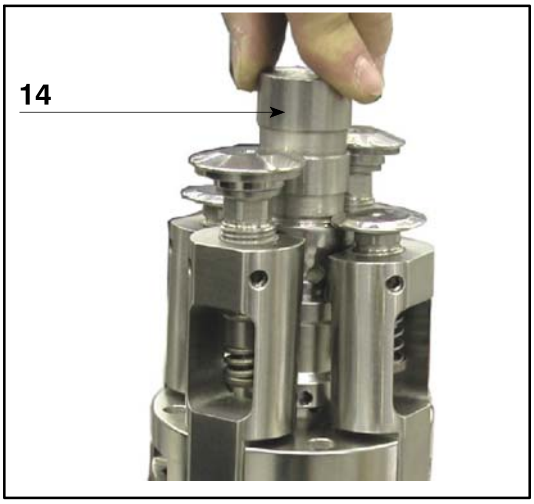


Fig.42

1.3.2. THREADING ROLLER ADJUSTMENT

- Using the wrench (**11**), loosen the grub screws (**H**) in such a way to free the bush (**I**);

Fig.43

- using the wrench (**8**) screw or unscrew the bush (**I**) in such a way to take the threading roller (**L**) to the height of the adjustment caliper plane;
- further screw or unscrew the bush (**I**) as indicated in the adjustment table;
- when the adjustment is complete, using the wrench (**11**), lock the grub screws (**H**);

Fig.44

Fig.45

- loosen the nut (**M**) using the wrench (**10**);
- using the wrench (**11**), loosen the grub screw (**N**) in such a way to take the threading roller (**L**) in contact with the caliper seat;
- lock the nut (**M**) making sure that the grub screw (**N**) does not move;
- repeat the same operation on all the threading arms;
- free the cam (**O**) taking it in rest position (upwards).

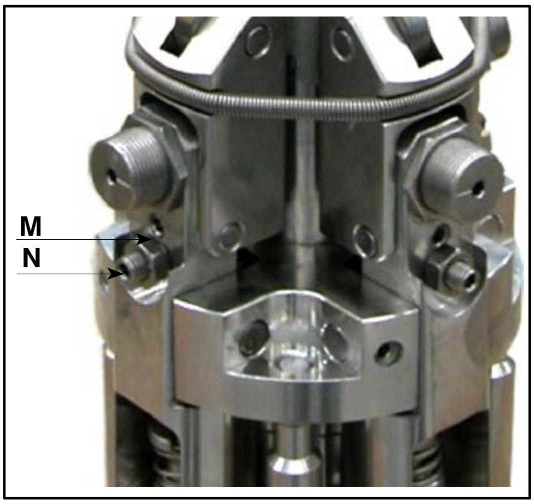


Fig.46

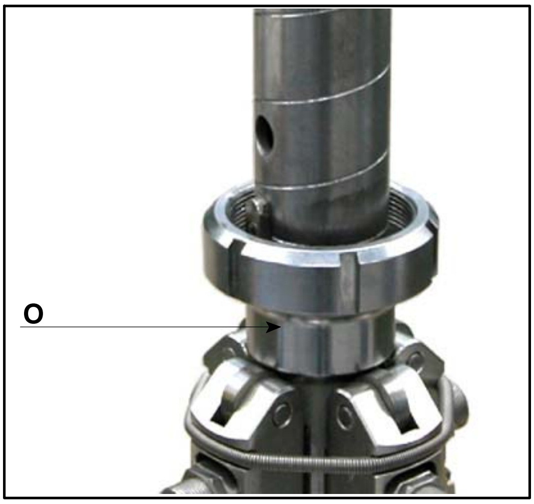


Fig.47

1.3.3. THREADING ROLLER LATERAL LOAD ADJUSTMENT

When the thread of the cap is too much pressed and the metal is cut in some points, it is necessary to:

- using the wrench (**9**) loosen the nut (**P**).
- using the wrench (**6**) unscrew the spring holder bush (**Q**) until obtaining the suitable engravement.

When the thread of the cap is not pressed, it is necessary to:

- using the wrench (**9**) loosen the nut (**P**).
- using the wrench (**11**) screw the spring holder bush (**Q**) until obtaining the suitable engravement.
- repeat the same operations for all the threading rollers.

<u>It is recommended, before starting the production, to carry out some closure tests.</u>

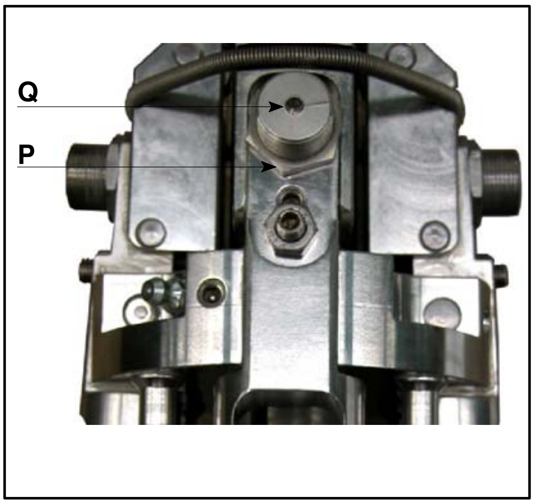


Fig.48

1.3.4. SEAMING ROLLER ADJUSTMENT

- Using the wrench (**11**), loosen the grub screws (**R**) in such a way to free the bush (**S**);

Fig.49

- using the wrench (**8**) screw or unscrew the bush (**S**) in such a way to take the seaming roller (**T**) to the height of the adjustment caliper (**14**) plane;
- further screw or unscrew the bush (**S**) as indicated in the adjustment table;
- when the adjustment is complete, using the wrench (**11**), lock the grub screws (**R**);

Fig.50

Fig.51

- loosen the nut (**U**) using the wrench (**10**);
- using the wrench (**11**), loosen the grub screw (**V**) in such a way to take the seaming roller (**T**) in contact with the caliper seat;
- lock the nut (**U**) making sure that the grub screw (**V**) does not move;

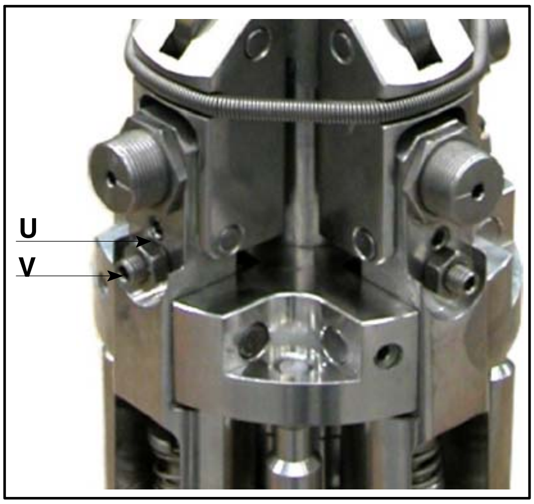


Fig.52

- repeat the same operation on all the seaming arms;
- free the cam (**M**) taking it in rest position (upwards).

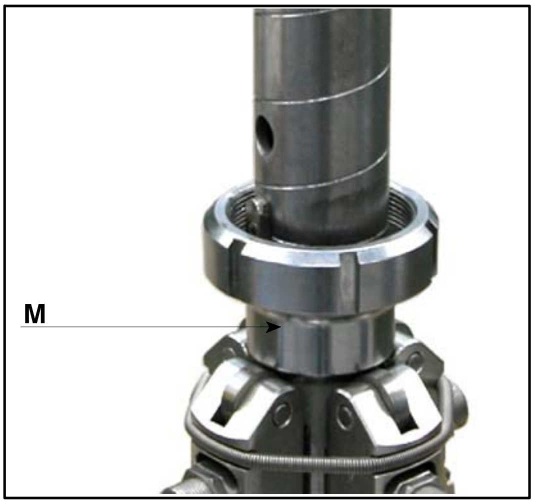


Fig.53

1.3.5. SEAMING ROLLER LATERAL LOAD ADJUSTMENT

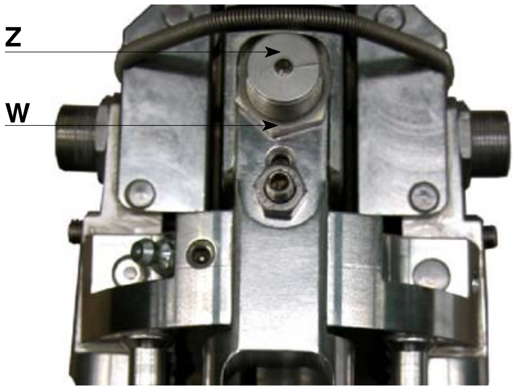


Fig.54

When the seaming on the cap is not well carried out, it is necessary:

- using the wrench (**9**) loosen the nut (**W**).
- using the wrench (**11**) screw or unscrew the spring holder bush (**Z**) until obtaining the suitable seaming;
- repeat the same operation on all the seaming rollers;

<u>It is recommended, before starting the production, to carry out some closure tests.</u>

1.4. MOUTHPRESSER REPLACEMENT

Remove the head from the machine as described in paragraph 1.2. on page 63 and operate as follows:

- insert the head in a jaw with soft claws;
- using the wrench (**11**), unscrew the fixing Allen screws (**E**) and remove the centering device (**F**);

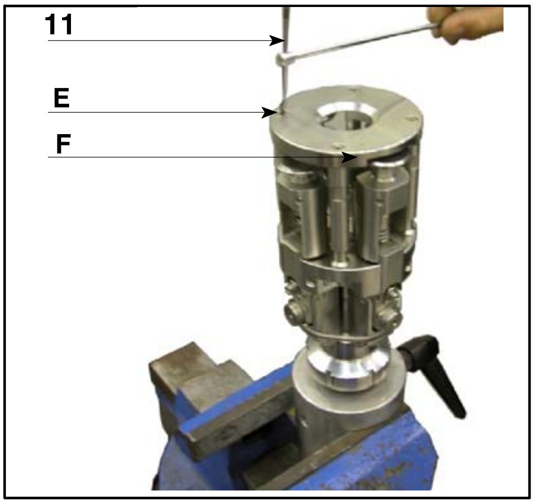


Fig.55

- using the wrench (**10**), unscrew and remove the studs (**G**) of the centering device.

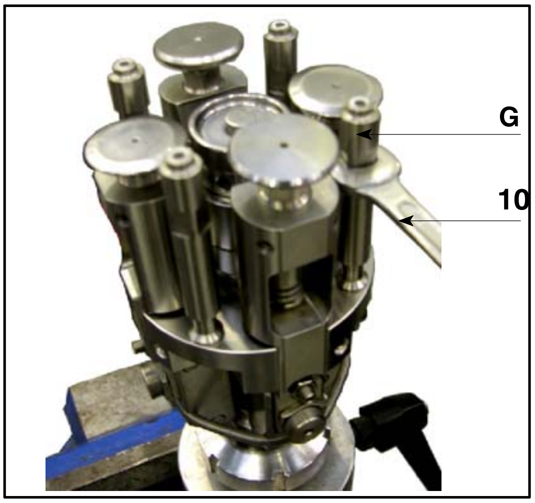


Fig.56

- To remove the mouthpresser it is necessary:
  - to remove the arm return spring (**J**);
  - using the wrench (**11**), loosen the grub screws (**K**);
  - slide the pins (**Y**) out;
  - remove the rollers (**W1**) and (**W2**) holder arms out;

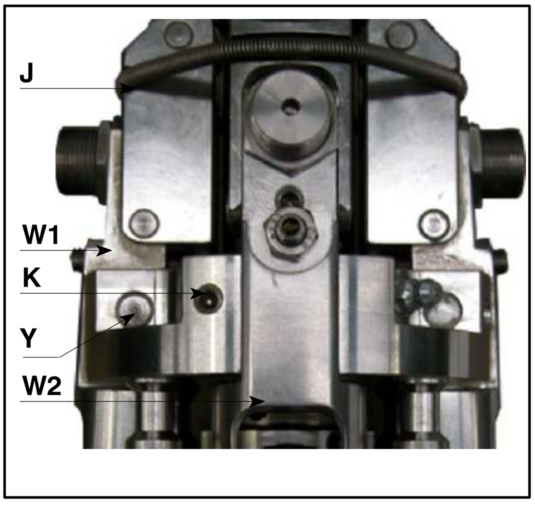


Fig.57

- Using the wrenches (**7**) inserted into the hole of the mouthpresser and in the hole of the central shaft (**AA**), unscrew and remove the mouthpresser (**AB**). If the mouthpresser is locked, it is sufficient to give a light stroke on the handle of the wrench.

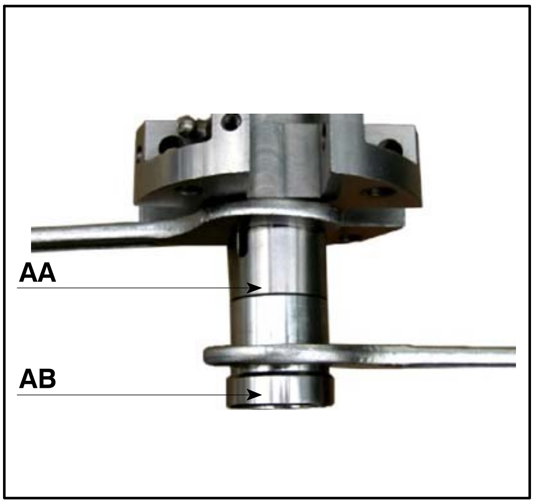


Fig.58

The mouthpresser threading is counterclockwise.

- Mount the mouthpresser for the new type of closure.
- Remount the arms and the centering device proceeding in the reverse way to what described before.
- Carry out the adjustment indicated in paragraph 1.3. on page 64.
- Remount the head.

1.5. SPRING REPLACEMENT

1.5.1. THREADING ROLLER SPRINGS

- Press the spring and slide the threading roller washer (**AC**) out from its seat.

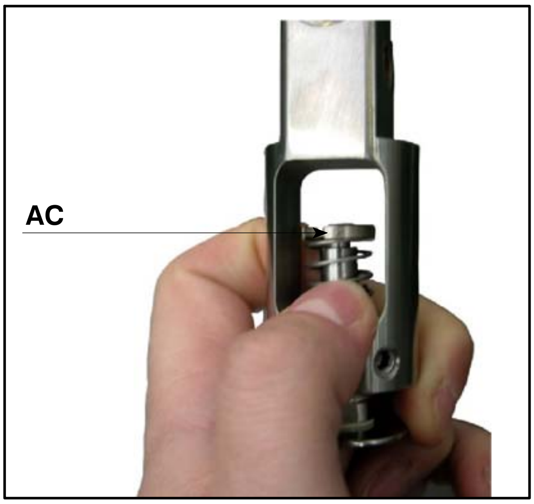


Fig.59

- Slide the spring (**AD**) out.
- Insert the new spring on the threading roller and keep it pressed.
- Reposition the threading roller washer (**AC**) in its seat.

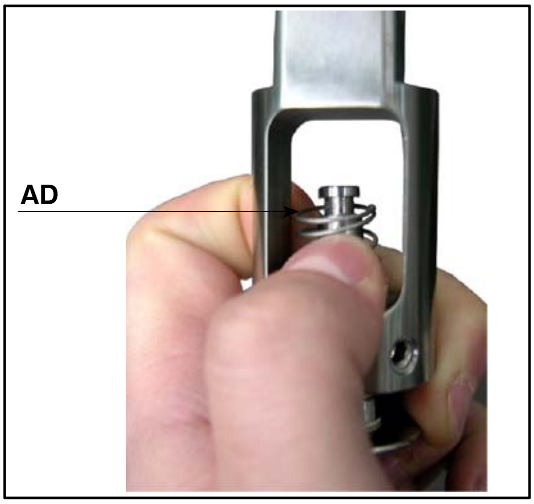


Fig.60

1.5.2. SEAMING ROLLER SPRING

- Push the spring (**AE**) downwards.
- Slide the seaming roller washer (**AF**) out of its seat.
- Slide the spring (**AE**) out.

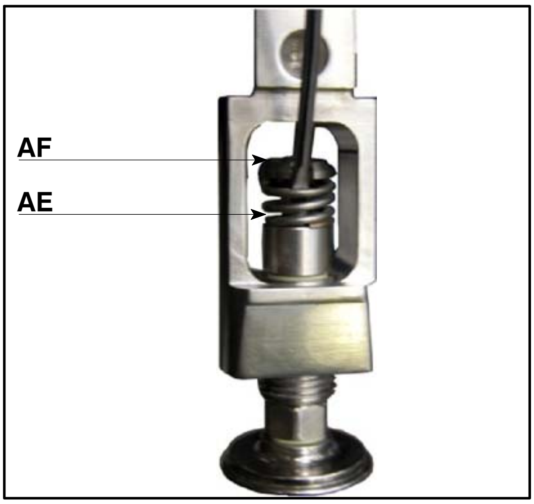


Fig.61

- Insert the roller holder arm (**W2**) complete with seaming roller (**T**) in a jaw with soft claws.
- Insert the new spring on the seaming roller and keep it pressed by the wrench (**13**).

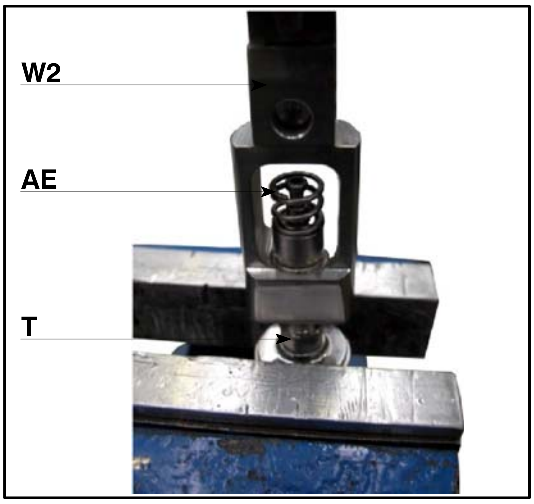


Fig.62

- Reposition the seaming roller washer (**AF**) in its seat.

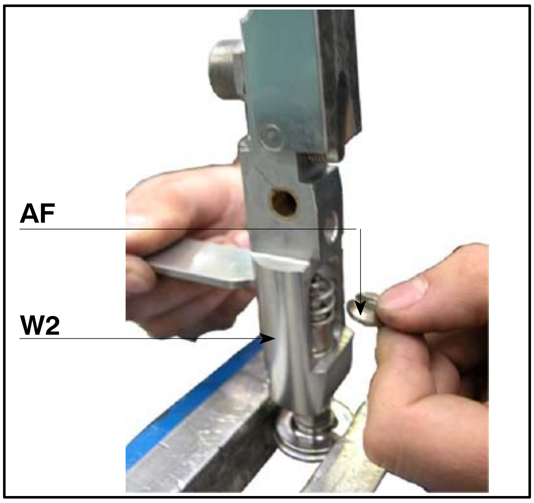


Fig.63

1.5.3. ARM RETURN SPRING

- Unhook the spring (**J**).
- Replace the spring with a new one.

1.5.4. LATERAL LOAD SPRINGS

- Using the wrench (**9**) loosen the nut (W).
- Using the wrench (**11**) unscrew the spring holder bush (**Z**).
- Slide the spring inside the bush out and replace it with a new one.
- Lightly grease the spring.
- Remount the bush (**Z**) and adjust the roller lateral load.
- Lock the nut (**W**).

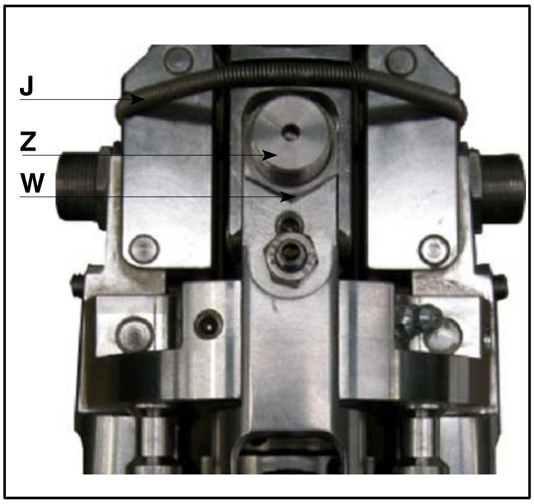


Fig.64

1.5.5. COMPENSATION SPRING

- Dismount the head from the spindle.
- Remove and replace the compensation spring (**AG**).
- Remount the head on the spindle.

Fig.65

1.6. HEAD CLEANING

It is recommended to use rags wetted with preferably warm water and non–aggressive detergents for the general cleaning.

The customer shall refer to the instructions by the supplier of the detergents used for cleaning the machine.

It is absolutely forbidden to wash with pressurized water jets or aggressive detergents.

<ins>Do not use cotton wool or other flimsy materials.</ins>

<ins>It is hazardous to use compressed air.</ins>

1.7. HEAD COMPONENT LUBRICATION

For the lubricant reference, see paragraph 3. on page 42

Fig.66

1.7.1. EVERY 8 HOURS

- Lubricate the arm rollers.
  Use **"I"**–type lubricant.

Fig.67

- Lubricate the threading and seaming rollers and springs.
  Use **"I"**–type lubricant.

1.7.2. EVERY 120 HOURS

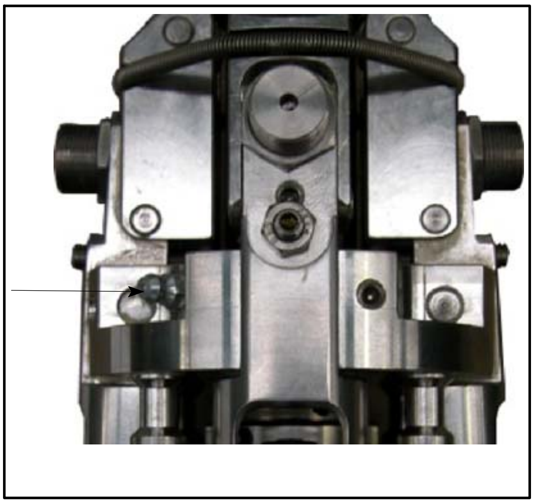


Fig.68

- Lubricate the head central block.
  Use **"K"**–type lubricant.

2. CENTRIFUGAL ORIENTATOR

2.1. INSTALLATION OF THE ORIENTATOR

For the installation of the centrifugal orientator see the listed below operations:

- remove the cover (**A**);

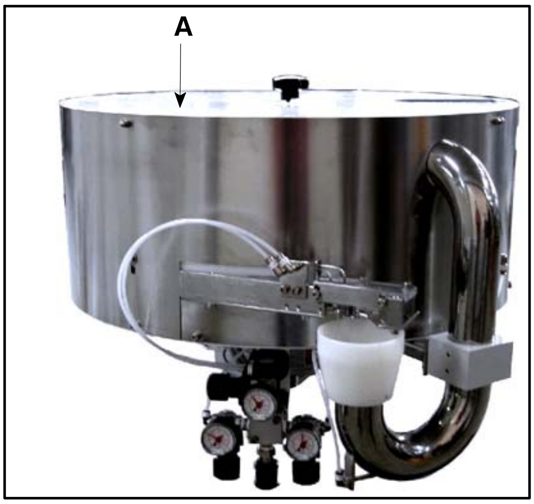


Fig.69

- remove screws (**B**) and unthread the central cone (**C**);

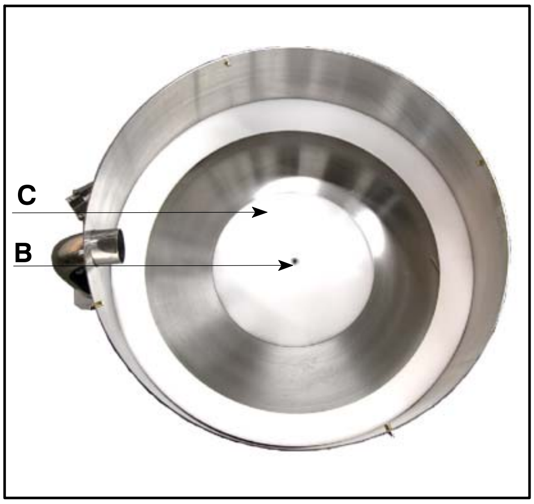


Fig.70

- screw at the center of the rotating disc an eyebolt (**D**);

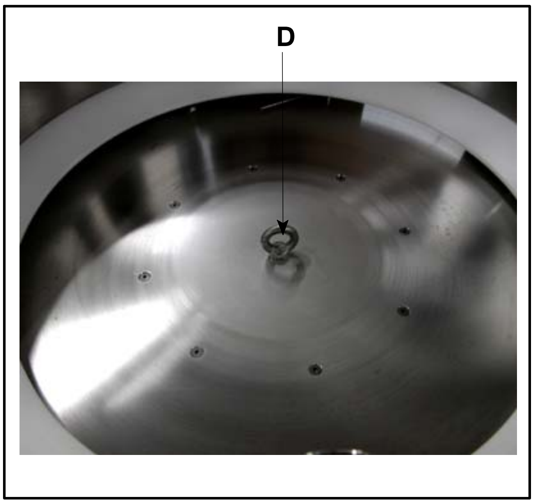


Fig.71

- hook with a belt or a chain with hooks (**E**) to the eyebolt (**D**) and lift the orientator with adeguated lifting means;

Do not stand or walk under the orientator when it is hoisted!

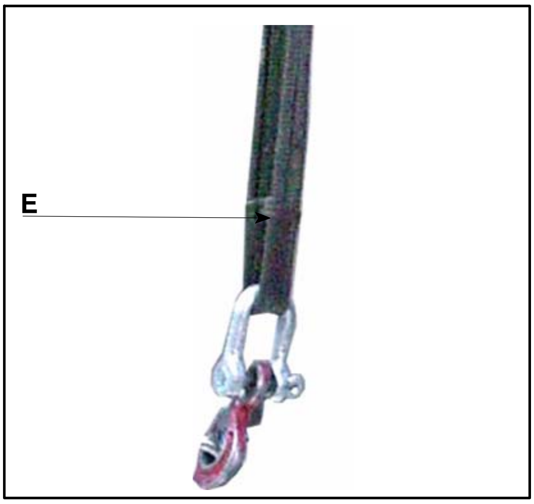


Fig.72

- place the orientator on the support (**F**), in such way that the exit mouth coincides with the caps chute;
- fix the orientator to the machine by bolts (**I**);
- remove the eyebolt (**D**), remount the cone and fix it with the screw (**B**);

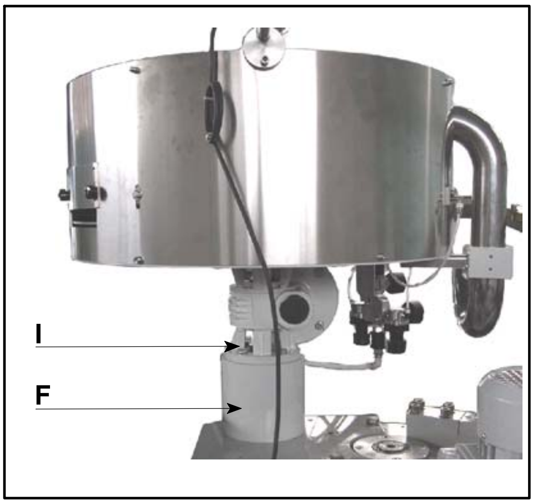


Fig.73

- if necessary, fix the caps chute to the caps exit mouth (**H**);

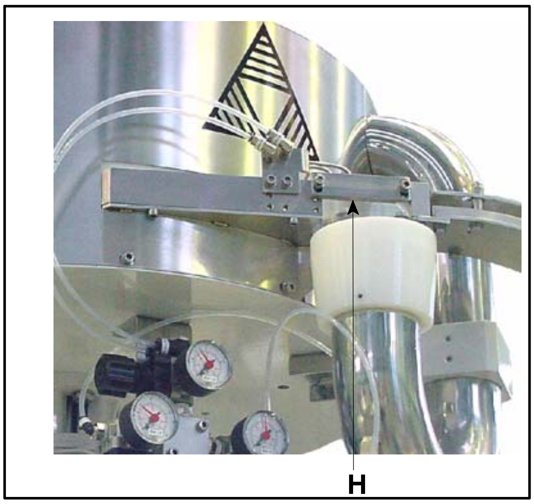


Fig.74

- connect the air main tube (ø8) of the main pneumatic unit to the coupling (**T**) of the pneumatic unit;
- if necessary, open the sliding valve (**U**);

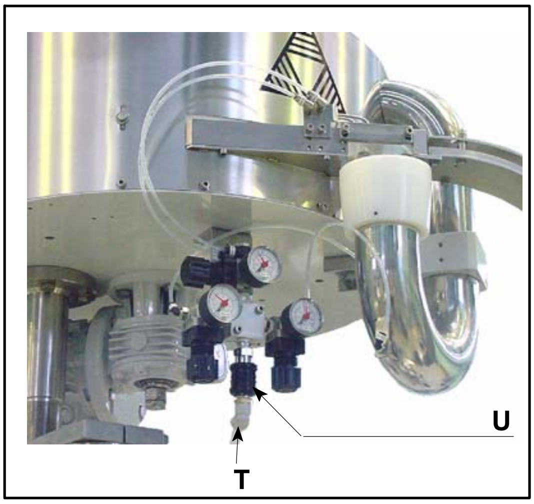


Fig.75

- connect the motor electric feeding cables;
- check that the rotation direction of the inner disc (**Z**) is correct; (the rotation must be sufficient in order to insert caps into the exit mouth), on the contrary:
  - invert 2 of the phases of motor feeding;
  - repeat the listed above operations to check if the rotation is regular.

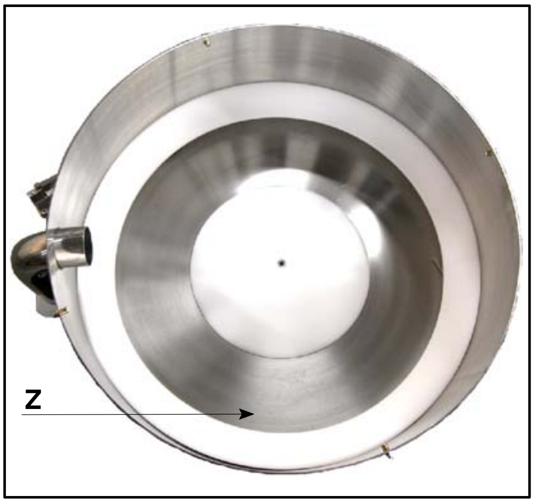


Fig.76

2.2. ORIENTATOR LOADING

For the loading of the orientator it is necessary to follow the precautions indicated in paragraph *"PRECAUTIONS FOR OPERATORS SAFETY"* in section *"GENERAL INFORMATION"*.

On the orientator is installed a caps level sensor (**L**) having the task to signal to the automatic elevator when it is necessary to put a new quantity of caps.

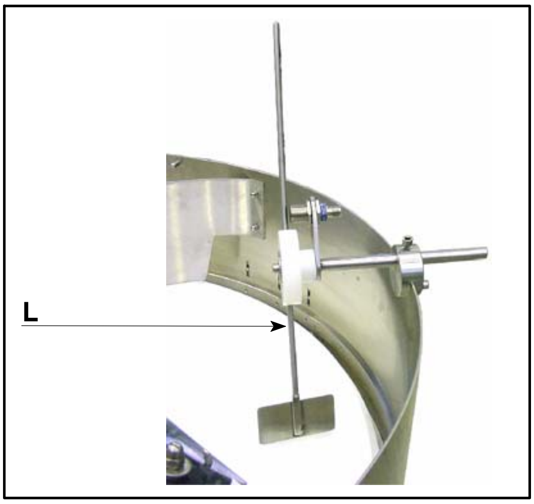


Fig.77

For a correct operating, if the orientator is loaded with different system from the automatic elevator, it is necessary to supply a quantity of caps that arrives at least 5 mm under the lower plate of the device.

2.3. ORIENTATOR EMPTYING

To discharge the orientator it is necessary to carry out the following operations:

- make sure the mode switch *AUT*/*MAN* is on *MAN* and remove the key.
- rotate the caps sorter until the caps orientator is totally empty;

2.4. CAPS FORMAT CHANGE

For the caps format change, it could be necessary to proceed as described below:

- remove the pneumatic connections (**M**);

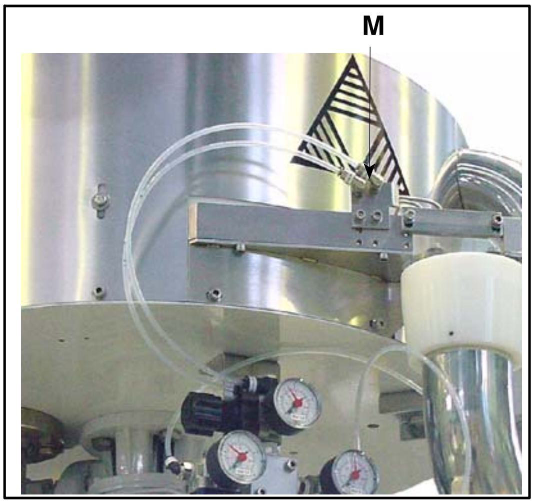


Fig.78

- remove the fixing screws (**N**),
- remove the whole mouth (**H**);

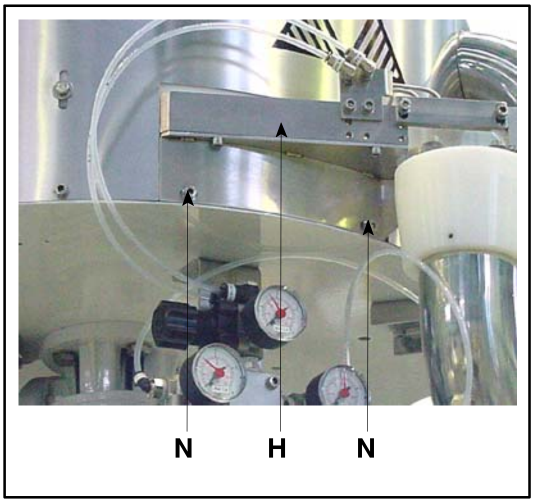


Fig.79

- if in the "set" there is also the return pipe (**O**) it is necessary to replace it;

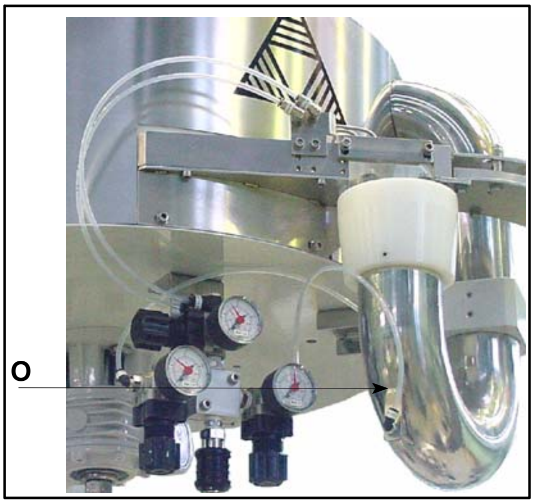


Fig.80

- if the height of the cap is different, adjust the height of the inner ring (**V**) carrying out on the screws (**X**) in such way that between the cap and the inner ring there is a space of 3 mm.;

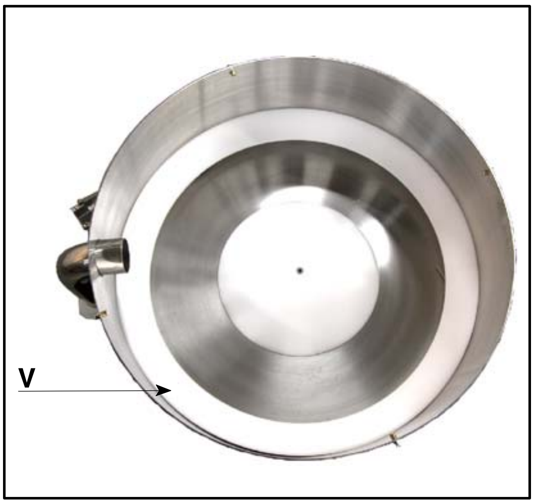


Fig.81

Fig.82

- on the orientator there is a door (**S**) that allows, thanks to the centrifugal force, the ejection of caps without seaming and seaming on them. If the height of caps to handle is different, it is necessary to adjust the position of the door carrying out on knobs (**K**), in such way that the caps without seaming can be ejected

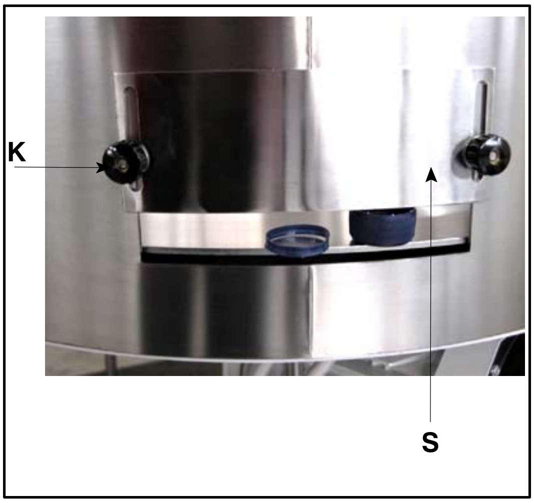


Fig.83

2.5. ADJUSTMENT OF THE ORIENTATOR

If the cap is not correctly shifted by the nozzles it is sufficient to increase or decrease the air pressure by the fitting pressure reducers.

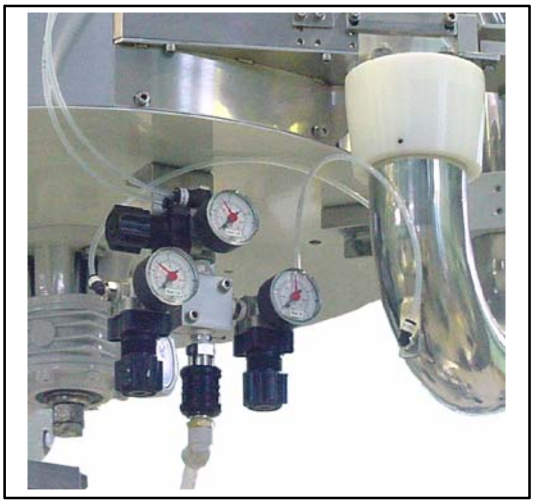


Fig.84

Use compressed air without impurities.

2.6. MAINTENANCE OF THE ORIENTATOR

The orientator does not need a particulary maintenance, it is sufficient a daily cleaning and the removal of deformed caps that may cause a caps chute obstruction .

If there is an anomalous distribution, it is necessary to carry out the following operations:

- check that the nozzles (**P**) are not obstructed and are in the correct position to push caps, not correctly orientated, into the caps chute;

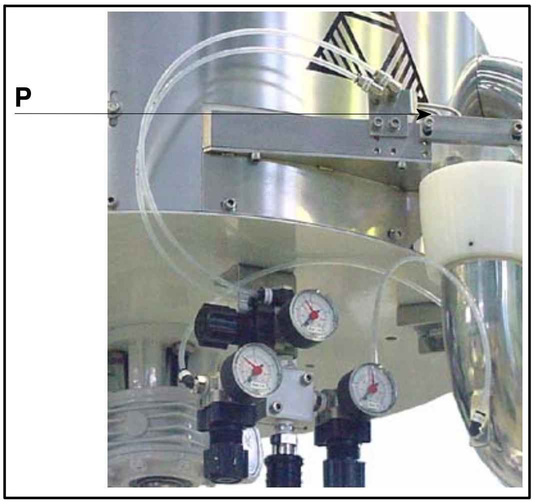


Fig.85

- check that the nozzle or nozzles, if more than one, (**Q**) is not obstructed;

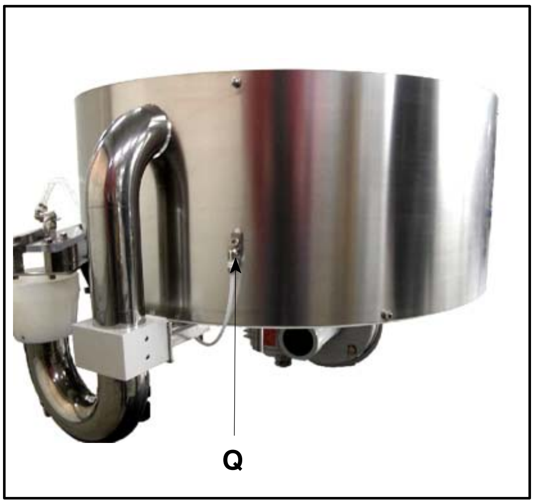


Fig.86

- check that the nozzle (**R**) is not obstructed;
- check that in the return pipe (**S**) there are no caps; if there are caps it is necessary to remove them.

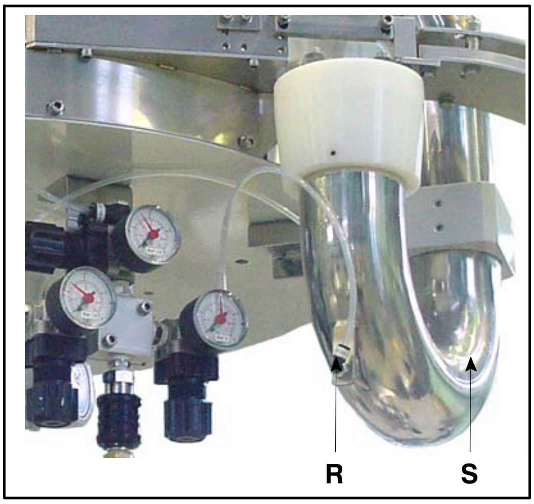


Fig.87

<u>Use compressed air without impurities.</u>

3. CAPS CHUTE

3.1. CAPS CHUTE INSTALLATION

To install the caps chute on the machine, it is necessary to:

- place the bracket (**A**) complete with the caps chute and caps distribution head on the bracket (**B**) of the upper plate;

<u>on the bracket (**B**) there are two centering dowels</u>

- fix by bolt (**C**);

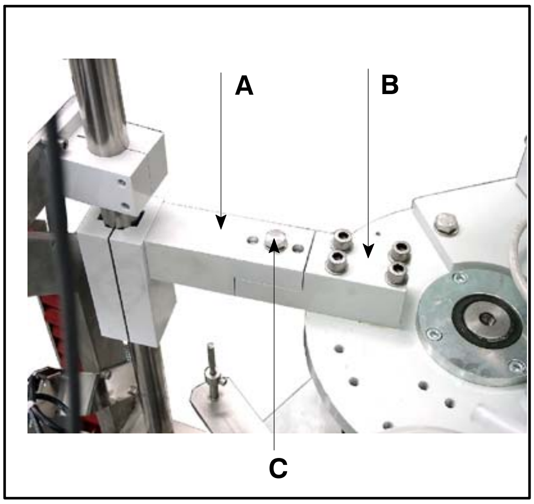


Fig.88

- check that the bottle neck is at the center of the caps distribution head (**D**);
- couple the electric connection of the photocell (**E**) (one or more).

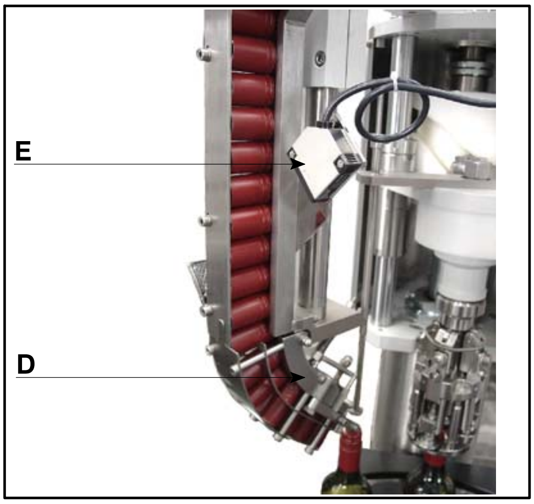


Fig.89

3.2. DISCHARGE OF THE CAPS CHUTE

To discharge the caps chute it is necessary to operate as follows:

- put a collect container under the the caps distribution head (**D**);
- remove the arms spring (**M**);
- open the arms (**N**);
- empty the caps chute;
- close the arms (**N**);
- reposition the arms spring (**M**);

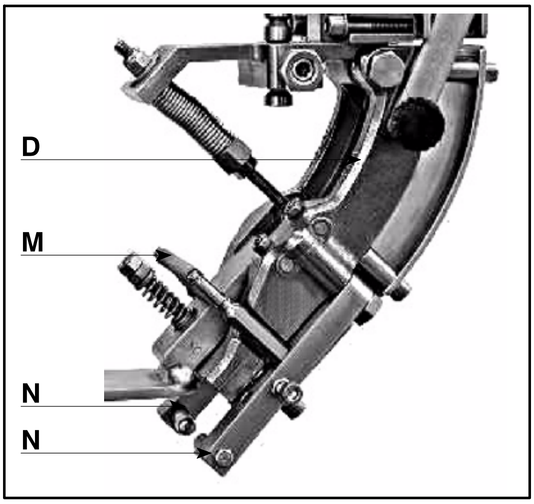


Fig.90

3.3. CAPS FORMAT CHANGE

When the cap to handle changes, it could be necessary to replace the caps chute and the distribution head proceeding as follows:

- disconnect the electric connections from the photocell (**E**) (one or more),

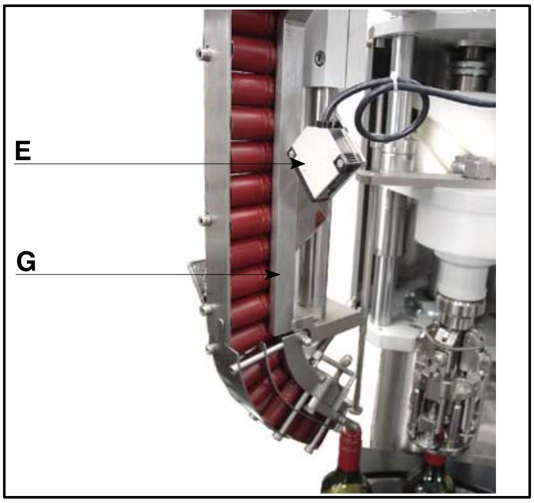


Fig.91

- unscrew and remove the bolt (**C**),
- remove the whole caps chute (**G**),
- mount, proceeding in the reverse way, the caps distribution unit suitable to the new type of cap to handle,

<u>on the bracket (**B**) there are two centering dowels</u>

- check that the caps chute is aligned to the caps exit mouth of the orientator
- connect the electric connections of the photocell (**E**) (one or more).

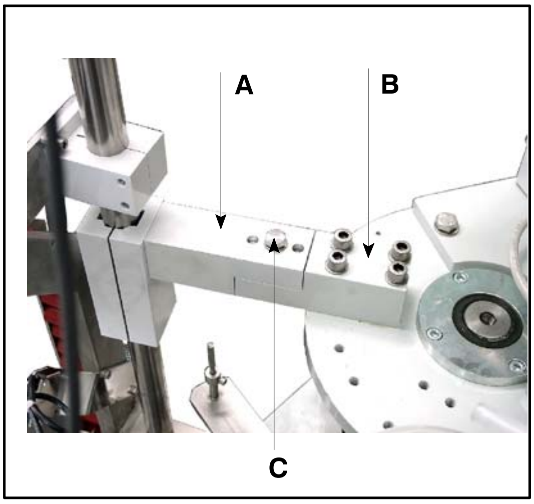


Fig.92

K PROBLEMS AND SOLUTIONS

1. INTRODUCTION

The problem list, in the following pages, can be helpful to find as soon as possible the causes of problems and how to eliminate them.

It is necessary that the finding and interventions on problems are carry out with method (from the most probable cause to the least probable cause) and following a logic sequence.

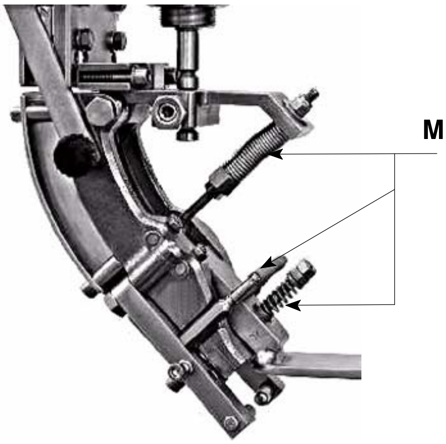
Follow the scale method: a first intervention, test of the obtained result and then passage to the following problem.

The list is referes to the machine. If the machine is working in sychron-ism with a production line, it is necessary to read the Manuals of the other machines of the line.

The main units of the machine are the following:

- containers transfer
- closure unit
- caps/corks distribution
- caps/corks feeding

PROBLEM	*CAUSE*	*SOLUTION*
Lifting or lowering the head, it does not move in height	Interferences with the containers conveying equipment, foreign bodies into the lifting mechanism, other things	Find the cause that prevents the lifting and elimite it
Containers are deformed or broken or are badly driven during the load phase	The star–wheel is not timed with the turret.	Time the star–wheel with the turret.
	Equipments for the containers trasfer are not suitable.	Replace equipement with the suitable one for the format to handle.
Containers are deformed or break during the closure phase	The head is too low	Place the head at the correct height-for the container to handle.
	Equipments for the containers trasfer are not suitable.	Replace the equipment with the suitable one to the format to handle.
Caps do not descent from the caps sorter	The exit mouth of the caps sorter is obstructed.	Remove any caps or part of caps or foreign bodies
		For any adjustment see the relevant paragraph.
	Off size or deformed caps.	Use caps of of the suitable format and with uniform characteristics.
The cap is too little or not threaded	Insufficient pressure of the threading rollers	Adjust the pressure of the threading rollers.
	Head height incorrect	Place the head at the correct height for the bottle to handle.
	The spring of the head is broken.	Dismount the head from the capping turret and replace the spring
	Locked threading rollers (seized up).	Manually check the rotation and the slide of threading rollers: If locked, dismount them, clean them and lubricate them.
	The spring of threading rollers is broken.	Replace the spring of the threading rollers.
The cap is too little or not seamed	Head of the seaming rollers is incorrect.	Adjust the height of the seaming rollers.
	Insufficient pressure of the seaming rollers.	Adjust the pressure of the seaming rollers
	The spring of the seaming rollers is broken	Replace the spring of the seaming rollers
The cap is cut on the thread	Excessive pressure of the threading rollers	Adjust the pressure of the threading rollers.
	Locked threading rollers (seized up).	Manually check the rotation and the slide of threading rollers: If locked, dismount them, clean them and lubricate them.

PROBLEM	***CAUSE***	***SOLUTION***
Problems in caps distribution	Incorrect cap gripper	Check the height of the head.
		Check the operating of springs (**M**) of the caps positioning device. 
		Check that the caps positioning device of caps is operating, If not contac the *Technical Assistance Service.*

Report of the training

By user care

TURRET AUTOMATIC SINGLE HEAD

model: ___________________________ **serial no.** ___________________________

manufacturing year:___________________________ **manufacturer:** ***CLOSYS S.r.l.***

On ______________ at the Company ____________________________ from ______________ to ______________ a training course concerning the use of turret has been held

Trainer/s:

Mr. ____

Mr. ____

Mr. ____

Mr. ____

The subject dealt with are the ones included in the "Use and maintenance" manual enclosed to the turret

A practical training on the turret: has been forseen **SI** **NO**

Attendees to the course:

NAME	CHARGE	SIGNATURE

The Attendee's signature implies the understanding of the dealt with subjects.

Trainer/s signature/s:

Mr. ____

Mr. ____

Mr. ____

Mr. ____